

Home-Foreign Elasticity Functions and the Gains from Trade*

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Abstract

Gains from trade are typically calculated using a constant trade elasticity estimated from substitution across foreign countries. For general homothetic preferences, we show how to calculate these gains from the initial domestic expenditure share and the home-foreign elasticity function along a price path to autarky. While this function generally depends on the specific price path, under weak separability it does not. We provide conditions for its nonparametric identification and estimate sector-level elasticity functions using tariff changes implied by trade agreements. Across WIOD and OECD ICIO data, estimated home-foreign elasticities generally rise with domestic expenditure shares. Using estimated elasticity functions in a multi-sector model with intermediate inputs, the median gain across the WIOD countries is 19.37 log points, compared with 24.43 when each country-sector elasticity is held fixed at its initial value. The corresponding increase in the mean is substantially larger, from 32.00 to 54.42 log points.

1 Introduction

How large are the welfare gains from international trade? In a broad class of quantitative trade models, the answer depends on two statistics: the domestic expenditure share and a trade elasticity that governs both substitution between home and foreign goods and substitution across foreign sources ([Arkolakis et al., 2012](#)). In practice, this elasticity is typically estimated from expenditure reallocation across foreign origins in response to changes in their relative prices. Thus, while the underlying theory restricts two distinct margins of substitution to be equally elastic, the empirical implementation identifies this common elasticity using variation only across foreign sources. This distinction matters: the welfare consequences of moving from trade to autarky generally depend on the home-foreign elasticity function, but not at all on the across-foreign-source elasticity that is used to identify it.

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The basic idea of this paper is simple. For general homothetic preferences, we show how gains from trade can be calculated along different price paths to autarky. A price path specifies how each foreign price changes relative to the home price as we move from the initial trade equilibrium toward autarky. Along this path, we measure the foreign relative-price change by averaging log changes in foreign prices relative to the home price, weighted by contemporaneous foreign expenditure shares. Under stated monotonicity conditions, gains from trade depend only on the initial domestic expenditure share and the home-foreign elasticity function along this path. Although every path yields the same gains from trade, the elasticity function can differ across paths. Without further restrictions, identifying this function from price variation along one path requires observing responses along that individual path all the way to autarky; even exogenous price changes along other paths will generally not identify the same function.

We therefore impose weak separability between home and foreign goods so that price changes along different paths can be used to identify the same home-foreign elasticity function. Together with homotheticity, weak separability makes the home-foreign expenditure ratio depend on foreign prices only through a foreign price index relative to the home price: changes in individual foreign prices matter only through their effect on this relative index. Hence, the home-foreign elasticity at a given domestic expenditure share is the same regardless of the price path used to reach that share. This allows us to combine responses to price changes along different paths to estimate the elasticity function used in the gains-from-trade calculation. Our general welfare results do not require this restriction; we impose it to identify the elasticity function using the variation available in the data.

For identification, we introduce multiple sectors, imposing the preceding structure within each sector and weak separability across sectors. The latter allows us to study each within-sector response independently of demand in other sectors. We initially treat each sector's home-foreign elasticity as an arbitrary function of its domestic expenditure share and assume that this function is common across countries within the sector; countries with different domestic expenditure shares can have different elasticities within the same sector. Cross-country variation in domestic expenditure shares and tariff changes then provides information about the sector's elasticity function. We also assume that tariff changes satisfy a conditional exclusion restriction. Under these assumptions, tariff variation identifies each sectoral elasticity function nonparametrically over the supported range of domestic expenditure shares.

For estimation, we approximate each sector's elasticity function with a restricted cubic spline. This function nests the standard constant-elasticity case as well as the case in which the elasticity is linear in the domestic expenditure share.¹ We allow intercepts to differ across sectors and pool the coefficients determining the shape of the function across sectors within broader sectoral groups. We consider three groupings: all goods-producing sectors together; manufacturing and non-manufacturing separately; and consumer manufacturing, producer manufacturing, and non-manufacturing separately.

We estimate these sector-level home-foreign elasticity functions directly, rather than inferring them from substitution across foreign sources. The dependent variable is the change within a given destination in expenditure on the home sector relative to aggregate foreign

¹Appendices C, D, and E provide fully specified quantitative Armington, Krugman, and Melitz models that generate the linear case.

expenditure in the same sector. Import-tariff changes shift the price of the foreign aggregate relative to the home price. We instrument tariff changes using changes implied by trade agreements and control for observed country-sector output-price changes. We allow the coefficients on these measured-price controls to differ from the home-foreign elasticities because output prices may be correlated with demand changes and the published indices are proxies for the prices in the model.

The baseline estimator requires the agreement instruments, after removing their linear relationship with the included controls, to be uncorrelated with the regression error, which includes unobserved demand changes. Under this exclusion restriction and the corresponding relevance condition, agreement-induced tariff variation identifies the level of the home-foreign elasticity, while its interactions with the initial domestic expenditure share and its spline term identify how that elasticity varies with the share. We estimate the resulting equation by two-stage least squares.

Because sectoral domestic expenditure is constructed under many assumptions and measured less directly than international trade flows, we implement this strategy in two independently constructed global data sets. Our baseline estimation uses WIOD for 1995 and 2007. It contains 627 destination-sector observations across 40 countries and 16 goods-producing sectors. We repeat the estimation using the OECD ICIO for the same years, which contains 2,030 observations across 80 countries and 26 goods-producing sectors. Both versions use tariff changes from the Global Tariff Database and industry output-price changes published alongside the corresponding input-output accounts. Both data sets measure total absorption, including intermediate and final uses. We therefore assume that final and intermediate users purchase the same sectoral composite and consequently have the same origin shares and home-foreign elasticity, as in, e.g., [Caliendo and Parro \(2015\)](#).

With sector-specific intercepts and a common spline shape across all goods-producing sectors, the average fitted home-foreign elasticity in WIOD rises from 3.33 at the sector-specific 25th percentiles of the domestic expenditure share to 7.01 at the 75th percentiles. The average slope evaluated at sector medians of the domestic expenditure share is 12.02. The corresponding OECD ICIO estimates are 2.14 and 6.89 at the 25th and 75th percentiles, with an average slope evaluated at sector medians of 11.72. In both data sets, the slope's 95 percent confidence interval based on standard errors clustered by destination country excludes zero. When the functions are allowed to differ across consumer manufacturing, producer manufacturing, and non-manufacturing goods, all average slopes remain positive in both data sets, although some are imprecisely estimated.

We subject these findings to an extensive sensitivity analysis that varies samples, pooling restrictions, tariff construction, functional form, influential observations, and identifying variation. These exercises include specifications that allow the level of each sector's elasticity function to vary with observable destination-country characteristics, replace the spline with linear or quadratic functions, and use lagged domestic expenditure shares to address potentially correlated measurement error. The conclusion that home-foreign elasticities generally increase with domestic expenditure shares is stable across these exercises.

We also use Monte Carlo exercises to assess the estimator's performance when the true elasticity functions are known. Using the quantitative multi-sector Armington model developed in [Appendix C](#), we consider both constant and increasing linear elasticity functions. The estimator produces slopes centered near zero when the true functions are constant and

accurately recovers positive slopes when the true functions are increasing and linear.

Finally, we combine the estimated elasticity functions with alternative assumptions about substitution across sectors and a parsimonious structure for intermediate inputs to calculate gains from trade. The unrestricted estimates yield negative home-foreign elasticities at some domestic expenditure shares. For quantification, we therefore use estimates that impose a positive lower bound on the full elasticity functions and replace the local tariff approximation with its finite-change counterpart. Imposing the bound reduces the magnitudes of the estimated slopes, but in WIOD their averages evaluated at sector medians of the domestic expenditure share remain positive in every group.

Allowing the home-foreign elasticity to vary introduces a connection between trade exposure and substitution along the path to autarky. For a given increasing elasticity function, a sector with a lower initial domestic expenditure share is not only more exposed to trade, but also begins at a lower elasticity. Moving to autarky therefore requires traveling a longer portion of the function, including a range of low elasticities at which replacing foreign expenditure with domestic expenditure is particularly costly. Holding the elasticity fixed at its initial value instead applies that low elasticity throughout the adjustment, producing larger gains from trade. To assess the quantitative importance of allowing elasticities to vary, we compare our baseline quantitative gains with a calculation that holds each country-sector elasticity fixed at its fitted initial value, leaving the other quantification assumptions unchanged.²

In WIOD, allowing elasticities to vary yields median gains from trade of 19.37 log points and mean gains of 32.00 log points, where gains are the log change in real income from autarky to the trade equilibrium. Holding elasticities fixed raises the median to 24.43 and the mean to 54.42 log points. The OECD ICIO results show the same pattern. Although the constrained estimated elasticity functions are not everywhere increasing, holding elasticities fixed raises gains for 39 of the 40 countries in the quantification using WIOD data and every country in the quantification using OECD ICIO data. Thus, allowing elasticities to change along the path to autarky substantially affects the measured gains even when the comparison holds their initial values fixed.

This paper contributes to the literature on sufficient statistics for the gains from trade. [Arkolakis et al. \(2012\)](#) show that, within a broad class of quantitative trade models, welfare changes can be recovered from changes in domestic expenditure shares and a constant trade elasticity. They recommend estimating this elasticity using the relationship between bilateral imports and trade costs, which identifies substitution across foreign sources.³ [Costinot and Rodríguez-Clare \(2018\)](#) characterize the welfare-relevant object more generally as the area under the compensated demand curve for foreign factor services. They also observe that foreign goods form a Hicks composite when their prices move proportionally relative to the home price. We formalize this insight and extend it to general price paths, showing that the home-foreign elasticity function depends on the particular path. We then provide restrictions

²This comparison retains differences in initial elasticities across destinations within each sector, unlike the usual benchmark with a common constant elasticity across destinations. Moreover, these elasticities are estimated from home-foreign substitution rather than substitution across foreign sources.

³[Ansari et al. \(2026\)](#) show that matching this gravity moment need not pin down gains from trade outside the maintained model: small departures from workhorse models can produce substantially different welfare predictions.

under which the same elasticity function applies across paths and can be estimated directly from home relative to foreign expenditure, rather than inferred from substitution across foreign sources.⁴

To connect our results to the quantitative trade models motivating the sufficient-statistics approach of [Arkolakis et al. \(2012\)](#), we develop fully specified Armington, Krugman, and Melitz-Pareto models in Appendices C, D, and E. These models generate home-foreign elasticity functions that are exactly linear in the domestic expenditure share. They nest standard CES demand while allowing the home-foreign elasticity both to vary with the domestic expenditure share and to differ from the elasticity across foreign sources.⁵ The Krugman extension yields the same gains-from-trade formula as the Armington model, showing that, within this parametric setting, the gains formula does not rely on the Armington supply structure. The Melitz-Pareto extension shows how the welfare calculation changes when firm selection also responds to trade costs.

The paper also contributes to the literature on nonparametric approaches to trade counterfactuals. [Adão et al. \(2017\)](#) show that counterfactual trade and welfare changes can be recovered from a reduced factor-demand system without imposing the functional-form restrictions of standard gravity models. Our objective is narrower: we seek to identify the home-foreign elasticity function needed to calculate gains from trade. Under weak separability, tariff changes that differ across foreign sources nonparametrically identify this function without requiring estimation of the full demand system across origins. Relatedly, [Adão et al. \(2026\)](#) estimate functions that summarize firm heterogeneity on the extensive and intensive export margins and use a monopolistic-competition model to quantify their implications for gains from trade. Whereas their approach estimates bilateral firm-export responses, we directly estimate the home-foreign elasticity function that enters the gains-from-trade calculation.

Finally, the paper also relates to the broader literature on trade elasticities. A large gravity literature estimates constant trade elasticities from bilateral variation across source countries (e.g., [Simonovska and Waugh, 2014](#); [Caliendo and Parro, 2015](#); [Boehm et al., 2023](#); [Teti, 2024](#)). We instead estimate substitution between the home good and all foreign sources jointly. This margin is also central to international macroeconomics, where it shapes trade-balance dynamics and the international transmission of shocks (e.g., [Backus et al., 1994](#); [Heathcote and Perri, 2002](#); [Corsetti et al., 2008](#)).⁶ Closest to our empirical analysis, [Feenstra et al. \(2018\)](#) distinguish the “macro” elasticity between home and imported goods from the “micro” elasticity across foreign sources, estimate both using detailed U.S. data, and use these estimates to calculate gains from trade. Like them, we distinguish between these two margins.⁷ We derive a gains-from-trade formula without imposing functional forms and

⁴Relatedly, [Baqae et al. \(2025\)](#) address the sensitivity of surplus measurement to demand-curve extrapolation by inferring firms’ consumer surplus from marginal-cost responses to supplier additions and separations. We instead estimate the home-foreign elasticity function and use it to integrate along the path to autarky.

⁵The multi-sector Armington model also provides the quantitative framework for our Monte Carlo exercises.

⁶Our estimated home-foreign elasticities vary across sectors and with domestic expenditure shares within sectors. Comparisons with constant trade elasticities therefore depend on which sectors and domestic expenditure shares are used to summarize our estimates.

⁷[Fajgelbaum et al. \(2020\)](#) and [Li et al. \(2022\)](#) also estimate distinct home-foreign and foreign-foreign

provide conditions for the home-foreign elasticity function's nonparametric identification. We allow this elasticity to vary with the domestic expenditure share and estimate its sector-specific function using variation across countries.

The remainder of the paper proceeds as follows. Section 2 derives the welfare results along general price paths. Section 3 introduces additional structure and provides a nonparametric identification result. Section 4 describes the empirical implementation and presents the elasticity estimates. Section 5 uses these estimates to quantify the gains from trade and Section 6 concludes.

2 Gains from Trade

Theoretical Framework. On the production side, the model is a standard Armington model of international trade (Armington, 1969). Let $\mathcal{N}$ denote a finite set of countries. Each country $i \in \mathcal{N}$ produces one country-specific differentiated good using an exogenously supplied and internationally immobile factor.⁸ Country i has an exogenous factor endowment $L_i > 0$ and linear production technology $y_i = a_i L_i$, with $a_i > 0$. Markets are competitive and each country has a representative consumer. Endogenous wages are w_i , so nominal income is $Y_i = w_i L_i$ and the unit production cost is $c_i = w_i/a_i$. Delivering one unit of country i 's good to country k requires shipping $\mathcal{D}_{ki} \geq 1$ units, with $\mathcal{D}_{ii} = 1$; $\mathcal{D}_{ki}$ is the iceberg cost of shipping. The consumer price in destination k is therefore $p_{ki} = \mathcal{D}_{ki} c_i$. Goods markets clear and trade is balanced.

Preferences, however, are more general. Country k has homothetic preferences over the vector $\mathbf{x}_k = (x_{ki})_{i \in \mathcal{N}}$, where $x_{ki} \geq 0$ denotes the quantity of country i 's good delivered to country k . These preferences are represented by a utility function u_k that is continuous, increasing, concave, and homogeneous of degree one. For strictly positive consumption bundles, u_k is twice continuously differentiable and strictly quasi-concave, and Hicksian demand is interior and continuously differentiable in prices at every strictly positive price vector.⁹ Country n 's preferences may contain arbitrary interactions between the domestic variety and foreign varieties.

Here, we focus on country n . Because u_n is homogeneous of degree one, define country n 's unit expenditure function as

$$e_n(\mathbf{p}_n) \equiv \min_{\mathbf{x}_n \geq 0} \left\{ \sum_{i \in \mathcal{N}} p_{ni} x_{ni} : u_n(\mathbf{x}_n) \geq 1 \right\}$$

elasticities and use them to study the welfare effects of trade policy. Melitz and Redding (2015) and Brooks and Pujolàs (2019) study the welfare implications of variable trade elasticities in models with firm selection and expenditure reallocation across sectors, respectively.

⁸The one-good-per-country formulation is for expositional convenience. Section 3 shows how the gains-from-trade result applies in a multi-sector environment. Appendices D and E replace the Armington supply side with, respectively, a Krugman model and a Melitz-Pareto model.

⁹Interiority is imposed only for tractability: if aggregate imports reach zero at a finite choke price, the welfare result can be re-stated over the range in which imports are positive, with autarky defined as the limiting endpoint.

and let

$$h_{ni}(\mathbf{p}_n) \equiv \frac{\partial e_n(\mathbf{p}_n)}{\partial p_{ni}}$$

denote the Hicksian demand corresponding to one unit of utility. The assumptions above imply that e_n is positive, increasing, concave, and homogeneous of degree one. By the assumed continuous differentiability of Hicksian demand and Shephard's lemma, e_n is twice continuously differentiable at every strictly positive price vector.

Let $X_{ni} \equiv p_{ni}x_{ni}$ denote expenditure by country n on goods from origin i , let $\lambda_{ni} \equiv X_{ni}/\sum_j X_{nj}$ denote its share of country n 's total expenditure, and define $X_{nF} \equiv \sum_{i \neq n} X_{ni}$ as n 's spending across all foreign sources. Country n 's expenditure shares sum to $1 - \lambda_{nn}$ across all foreign sources. Denote by $W_n \equiv Y_n/e_n(\mathbf{p}_n)$ country n 's real income.

Welfare Along General Price Paths. We first show how welfare changes can be calculated along different price paths under the general homothetic preferences above. We then briefly discuss why one particular price path—in which all foreign prices rise proportionally—requires less information.

Consider a continuously differentiable path Γ in country n 's consumer-price space, indexed by $t \geq 1$ and beginning at the consumer-price vector in the initial equilibrium, where $t = 1$. All quantities evaluated along this path depend on the choice of Γ . We suppress this dependence in the notation except for the home-foreign elasticity, where we retain the superscript Γ to emphasize that different paths can imply different elasticity functions. Let $p_{ni}(t)$ denote the consumer price in country n of the good produced by country i at point t along this particular path Γ , and define the normalized price vector by

$$\pi_{ni}(t) \equiv p_{ni}(t)/p_{nn}(t).$$

Define unit expenditure along the normalized path by

$$H_n(t) \equiv e_n(\boldsymbol{\pi}_n(t)),$$

where $H_n(t)$ is expressed in units of the domestic good: by construction, $\mathbf{p}_n(t) = p_{nn}(t)\boldsymbol{\pi}_n(t)$, so homogeneity implies $e_n(\mathbf{p}_n(t)) = p_{nn}(t)H_n(t)$. Let

$$\mu_{ni}(t) \equiv \frac{X_{ni}(t)}{X_{nF}(t)} = \frac{\lambda_{ni}(t)}{1 - \lambda_{nn}(t)}$$

denote origin i 's share of foreign expenditure at point t . Using these shares at each point along the path as weights, define the cumulative log relative-price change, $z_n(t)$, as

$$z_n(t) \equiv \int_1^t \sum_{i \neq n} \mu_{ni}(u) \frac{d \ln \pi_{ni}(u)}{du} du. \quad (1)$$

Thus, each foreign price change relative to the home price is weighted by that origin's share of foreign expenditure at that specific point along the path. Differentiating $H_n(t)$ and applying Shephard's lemma gives

$$d \ln H_n(t) = [1 - \lambda_{nn}(t)] dz_n(t). \quad (2)$$

The change in unit expenditure relative to the home price is therefore the foreign expenditure share times the weighted foreign relative-price change. Moreover, since $Y_n/p_{nn} = a_n L_n$, real income at each equilibrium endpoint is $W_n = a_n L_n/H_n$. The welfare comparison uses income at these endpoints; the intermediate points on the price path are used to evaluate the expenditure function.

Suppose $z'_n(t) > 0$ throughout the relevant interval; that is, the cumulative log relative-price change increases along this part of the path, even though individual foreign relative prices may fall. We can define the home-foreign elasticity along the particular path Γ as

$$\varepsilon_n^\Gamma(t) \equiv \frac{d \ln[X_{nn}(t)/X_{nF}(t)]/dt}{z'_n(t)}. \quad (3)$$

This is both the Hicksian and Marshallian elasticity of the home-to-foreign expenditure ratio along this path.¹⁰ The assumed continuous differentiability of Hicksian demand makes expenditure shares continuously differentiable, so the derivative in equation (3) is well defined. The superscript Γ indexes the price path along which the elasticity is measured.

Since $X_{nn}/X_{nF} = \lambda_{nn}/(1 - \lambda_{nn})$, the definition of $\varepsilon_n^\Gamma(t)$ yields

$$d \ln \frac{\lambda_{nn}(t)}{1 - \lambda_{nn}(t)} = \varepsilon_n^\Gamma(t) dz_n(t).$$

If, in addition, the elasticity is strictly positive throughout the interval, the domestic expenditure share is strictly increasing and can itself index the path. In this case, we write the elasticity function as $\varepsilon_n^\Gamma(\lambda) \equiv \varepsilon_n^\Gamma(t(\lambda))$. The next proposition gives welfare changes in the general case and an alternative expression indexed by the domestic expenditure share under these positivity conditions. Proofs of Proposition 1 and Corollary 1 are provided in Appendix B.

Proposition 1. *Consider two equilibria, indexed by 0 and 1, connected by the price path Γ , corresponding to $t = t_0$ and $t = t_1$, where $t_1 \geq t_0 \geq 1$. Hold country n 's preferences, technology, and factor endowment fixed. Then*

$$\ln \frac{W_n^1}{W_n^0} = - \int_{t_0}^{t_1} [1 - \lambda_{nn}(t)] z'_n(t) dt. \quad (4)$$

If, in addition, $z'_n(t) > 0$ and $\varepsilon_n^\Gamma(t) > 0$ along the portion of the path connecting the two equilibria, then

$$\ln \frac{W_n^1}{W_n^0} = \int_{\lambda_{nn}^1}^{\lambda_{nn}^0} \frac{d \ln \lambda}{\varepsilon_n^\Gamma(\lambda)}. \quad (5)$$

Equation (4) continues to apply when the weighted price change or the domestic expenditure share reverses direction. The additional positivity conditions allow the domestic share to index the path in equation (5).

¹⁰For any price vector $\mathbf{p}_n$, Hicksian demand at an arbitrary utility level U is $U \mathbf{h}_n(\mathbf{p}_n)$ and Marshallian demand at an arbitrary expenditure level Y_n is $[Y_n/e_n(\mathbf{p}_n)] \mathbf{h}_n(\mathbf{p}_n)$. In each case, the scalar multiplying $\mathbf{h}_n(\mathbf{p}_n)$ cancels from the ratio X_{nn}/X_{nF} . Hence, this ratio and its response along the price path are the same under Hicksian and Marshallian demand.

Gains from Trade. For a finite counterfactual from t_0 to $t_1 < \infty$, the terminal relative-price vector, $\pi_n(t_1)$, is itself an outcome to be determined. The autarky counterfactual is special because it corresponds to every foreign price relative to the home price tending to infinity. To focus on finite gains from trade, assume that foreign goods are not jointly essential: one unit of utility can be attained using a finite quantity of the home good alone.¹¹ This assumption implies that the domestic expenditure share converges to one as all foreign relative prices tend to infinity, without requiring a sign restriction on $\varepsilon_n^\Gamma(t)$.¹²

Corollary 1. *Let T denote an initial trade equilibrium at $t = 1$, and let A denote autarky for country n . Hold country n 's preferences, technology, and factor endowment fixed. Suppose one unit of utility can be attained using a finite quantity of the home good alone, and consider a price path Γ along which $\pi_{ni}(t) \rightarrow \infty$ as $t \rightarrow \infty$ for every foreign origin. Then $\lambda_{nn}(t) \rightarrow 1$, $W_n^A > 0$, and*

$$\ln \frac{W_n^T}{W_n^A} = \int_1^\infty [1 - \lambda_{nn}(t)] z'_n(t) dt. \quad (6)$$

If, in addition, $z'_n(t) > 0$ and $\varepsilon_n^\Gamma(t) > 0$ for every finite $t \geq 1$, then

$$\ln \frac{W_n^T}{W_n^A} = \int_{\lambda_{nn}^T}^1 \frac{d \ln \lambda}{\varepsilon_n^\Gamma(\lambda)}. \quad (7)$$

We define gains from trade as the log change in real income from autarky to the trade equilibrium. Every path satisfying the stated conditions gives the same gains from trade because the integral equals the change in log unit expenditure between the same endpoints. Under the positivity conditions, these gains can be calculated from the initial domestic expenditure share and the home-foreign elasticity function along the chosen path. The elasticity function can differ across paths even though its integral gives the same gains from trade.

The Proportional Foreign-Price Ray. Constructing this home-foreign elasticity function generally requires knowing how expenditure shares across foreign origins evolve along the path. These shares are needed to measure the weighted foreign relative-price change in equation (1). If we do not observe them directly, predicting them generally requires modeling substitution across foreign origins. With general homothetic preferences, this can involve a large matrix of cross-price elasticities at each point along the path.

However, Costinot and Rodríguez-Clare (2018) observe that foreign goods can be treated as a Hicks composite along a path on which their prices move proportionately relative to the domestic price. This path provides a particularly simple case of the results above. Let

¹¹This is a condition on the level of utility at the boundary of the consumption set where all foreign consumption is zero. It is distinct from standard Inada conditions, which restrict marginal utilities as consumption approaches the boundary. If this assumption is violated, the welfare formula below continues to apply along every finite portion of the path, but gains from trade are infinite.

¹²If this assumption is violated, equation (4) continues to apply on every finite portion of the path, but real income converges to zero as all foreign relative prices tend to infinity, so gains from trade are infinite. Cobb-Douglas preferences over the domestic good and a foreign aggregate illustrate this case: the domestic expenditure share is fixed at the Cobb-Douglas parameter and therefore does not converge to one.

$m \geq 1$ denote the common multiplier applied to every foreign price relative to the home price, so that

$$\pi_{ni}(m) = m \frac{p_{ni}(1)}{p_{nn}(1)} \quad \text{for every } i \neq n. \quad (8)$$

Since the foreign expenditure shares sum to one, the weighted price change equals the common change in $\ln m$, regardless of how these shares evolve. Hence, $z_n(m) = \ln m$ and the home-foreign elasticity along this ray is

$$\varepsilon_n(m) \equiv \frac{d \ln[X_{nn}(m)/X_{nF}(m)]}{d \ln m}.$$

Knowing the domestic expenditure share as a function of m is therefore sufficient to calculate gains from trade. We do not need to recover expenditure allocations across foreign origins along the ray. When $\varepsilon_n(m) > 0$ for every finite $m \geq 1$, this elasticity function can be indexed by the domestic expenditure share from λ_{nn}^T to one, as in Corollary 1. This simplifies the information needed to calculate gains from trade.

Unfortunately, it does not simplify the identification of the home-foreign trade elasticity. Identification requires that we know how the domestic expenditure share responds as all foreign prices rise proportionally relative to the home price. In practice, observed price changes simply do not follow this ray: they differ across foreign origins. Without further restrictions on demand, the responses to these non-proportional changes need not identify the elasticity function along the proportional ray.

3 Weak Separability and Identification

The general result in Section 2 allows us to calculate gains from trade along paths other than the proportional foreign-price ray. This flexibility does not, however, resolve the identification problem. Although every path yields the same gains from trade, the elasticity function can differ across paths. Without further restrictions, identifying this function from price variation along one path requires observing responses along that individual path all the way to autarky; even exogenous price changes along other paths will generally not identify the same function.

We therefore impose weak separability between home and foreign goods so that responses to price changes along different paths identify the same home-foreign elasticity function. We introduce this restriction within each sector of a multi-sector model, maintaining homothetic preferences at each nest. The home-foreign expenditure ratio then depends on foreign prices only through an exact foreign price index relative to the domestic price, whose log change equals the weighted relative-price change in Section 2. At a given domestic expenditure share, the home-foreign elasticity is therefore the same across price paths. We also impose weak separability across sectors, allowing us to study each within-sector response independently of demand in other sectors.

We derive the formula for evaluating gains from trade in this environment in Section 3.1. Section 3.2 then establishes conditions under which cross-country variation in tariffs and domestic expenditure shares identifies the sector-specific home-foreign elasticity functions that enter this formula.

3.1 Weak Separability and Gains from Trade

Let $s \in \mathcal{S}$ index sectors. Suppose preferences in country n are weakly separable across sectors, each sectoral sub-preference is homothetic, and the cross-sector aggregator is homothetic. Within each sector, suppose preferences are weakly separable between a home good and a foreign aggregate, with homothetic preferences both within the foreign aggregate and between the home good and foreign aggregate. At each nest, we maintain the regularity assumptions on preferences and demand stated in Section 2. Let P_{nFs} denote the foreign aggregate's price index, $e_{ns}(p_{nns}, P_{nFs})$ the sector- s unit expenditure function, and $\mathcal{E}_n((e_{ns})_{s \in \mathcal{S}})$ the cross-sector unit expenditure function. Under this nested homothetic structure, the shares of sectoral expenditure allocated to the home good and foreign aggregate depend only on the relative price P_{nFs}/p_{nns} . We can therefore study this within-sector allocation without specifying how total expenditure is allocated across sectors.

Let X_{nns} and X_{nFs} denote expenditure on the home good and foreign aggregate in sector s , let λ_{nns} denote the home expenditure share within sector s , and define the sectoral home-foreign elasticity in sector s as

$$\varepsilon_{ns}(\lambda_{nns}) \equiv \frac{\partial \ln(X_{nns}/X_{nFs})}{\partial \ln(P_{nFs}/p_{nns})},$$

which we assume is strictly positive throughout the relevant interval of sectoral domestic expenditure shares. This restriction allows the domestic expenditure share to index the relative-price path within each sector, as in the expenditure-share representation of Proposition 1. Weak separability between home and foreign goods allows us to treat demand within each sector as a choice between the home good and the foreign aggregate, whose price is P_{nFs} . Because the foreign sub-preference is homothetic, holding preferences fixed, Shephard's lemma gives

$$d \ln(P_{nFs}/p_{nns}) = \sum_{i \neq n} \mu_{nis} d \ln(p_{nis}/p_{nns}),$$

where μ_{nis} is origin i 's share of foreign expenditure in sector s . Thus, the weighted relative-price change in Section 2 is the differential of an exact foreign price index. Because the home-foreign expenditure ratio depends only on P_{nFs}/p_{nns} , the function $\varepsilon_{ns}(\lambda)$ is common across source-specific price paths. Let $H_{ns} \equiv e_{ns}/p_{nns}$ denote sectoral unit expenditure expressed in units of the domestic sectoral good; this is the sector-specific counterpart of H_n in Section 2. Applying the same argument within sector s gives¹³

$$d \ln H_{ns} = d \ln \left(\frac{e_{ns}}{p_{nns}} \right) = \frac{d \ln \lambda_{nns}}{\varepsilon_{ns}(\lambda_{nns})}. \quad (9)$$

Equation (9) determines the change in each sectoral price index relative to the corresponding domestic price. Combining these sectoral changes into aggregate welfare requires the cross-sector aggregator $\mathcal{E}_n$.

¹³The result follows from the same demand-side argument used in Proposition 1. For a two-good problem with home price p , foreign-aggregate price P , domestic expenditure share λ , and home-foreign elasticity $\varepsilon(\lambda)$, we have $d \ln(e/p) = (1 - \lambda)d \ln(P/p)$ and $d \ln[\lambda/(1 - \lambda)] = \varepsilon(\lambda)d \ln(P/p)$. Combining these expressions gives $d \ln(e/p) = d \ln \lambda / \varepsilon(\lambda)$.

For two equilibria, indexed by 0 and 1, across which the sectoral and cross-sector preference shifters are fixed, define $\Delta \ln H_{ns} \equiv \ln H_{ns}^1 - \ln H_{ns}^0$ as the change in log sectoral unit expenditure relative to the domestic sectoral price. Integrating equation (9) between the two equilibria gives

$$\Delta \ln H_{ns} = \int_{\lambda_{ns}^0}^{\lambda_{ns}^1} \frac{d \ln \lambda}{\varepsilon_{ns}(\lambda)}.$$

Equivalently, exponentiating the definition of $\Delta \ln H_{ns}$ yields

$$e_{ns}^1 = e_{ns}^0 \frac{p_{nns}^1}{p_{nns}^0} \exp(\Delta \ln H_{ns}).$$

Therefore, without imposing a particular cross-sector aggregator,

$$\ln \frac{W_n^0}{W_n^1} = \ln \frac{Y_n^0}{Y_n^1} + \ln \frac{\mathcal{E}_n((e_{ns}^0 (p_{nns}^1/p_{nns}^0) \exp(\Delta \ln H_{ns}))_{s \in \mathcal{S}})}{\mathcal{E}_n((e_{ns}^0)_{s \in \mathcal{S}})}. \quad (10)$$

Suppose nominal income and all domestic sectoral prices change proportionally, so that $Y_n^1/Y_n^0 = p_{nns}^1/p_{nns}^0 = q_n$ for every sector s . This condition holds in our one-factor Armington model when sectoral productivities are fixed across equilibria. Homogeneity permits dividing equilibrium 1 nominal income and all prices by q_n without changing real income. Because all domestic-sector prices change by q_n , this normalization returns them to their equilibrium 0 levels, while sector s 's unit expenditure becomes

$$\frac{e_{ns}^1}{q_n} = e_{ns}^0 \exp(\Delta \ln H_{ns}).$$

Hence, using homogeneity of $\mathcal{E}_n$, equation (10) becomes

$$\ln \frac{W_n^0}{W_n^1} = \ln \frac{\mathcal{E}_n((e_{ns}^0 \exp(\Delta \ln H_{ns}))_{s \in \mathcal{S}})}{\mathcal{E}_n((e_{ns}^0)_{s \in \mathcal{S}})}. \quad (11)$$

Under this proportionality condition, the aggregate gains-from-trade result in Section 2 also applies directly to this multi-sector economy. Let λ_n denote the share of total expenditure allocated to domestically produced goods across all sectors, and let $\varepsilon_n^\Gamma(\lambda)$ denote the elasticity of aggregate home relative to foreign expenditure along a price path Γ , defined using contemporaneous expenditure shares to weight price changes across all foreign goods. Under the conditions of Corollary 1, gains from trade equal $\int_{\lambda_n^T}^1 d \ln \lambda / \varepsilon_n^\Gamma(\lambda)$. Hence, this aggregate formula applies in the many-sector model and does not require weak separability within or across sectors. We impose these restrictions to identify sectoral elasticity functions that are independent of the price path and use them to calculate gains from trade.

Equation (11) expresses the welfare change in terms of sector-level changes in H_{ns} . For the trade-to-autarky comparison, equation (9) implies that $\Delta \ln H_{ns} = \int_{\lambda_{ns}^T}^1 d \ln \lambda / \varepsilon_{ns}(\lambda)$: in each sector, the domestic expenditure share rises from its trade value to one. Aggregating these sectoral changes, equation (11) shows that gains from trade also depend on the cross-sector aggregator $\mathcal{E}_n$.

As a familiar benchmark, suppose $\mathcal{E}_n$ is Cobb-Douglas with expenditure weights α_{ns} , as in many quantitative trade models (e.g., [Costinot and Rodríguez-Clare, 2018](#)). Then

$$\ln \frac{W_n^T}{W_n^A} = \sum_{s \in \mathcal{S}} \alpha_{ns} \int_{\lambda_{ns}^T}^1 \frac{d \ln \lambda}{\varepsilon_{ns}(\lambda)}. \quad (12)$$

The baseline quantification in Section 5 instead uses the more flexible nested CES cross-sector aggregator.

Common Sourcing Across Users. The demand system above is stated for a representative consumer. The data we use in estimation instead measure total absorption, which aggregates across household consumption, government purchases, investment, and intermediate-input demand. We aim to estimate the elasticity functions without separately modeling each user. To achieve this, we impose a common sourcing assumption within each destination-sector cell. Specifically, all users within a destination-sector cell face the same origin prices and allocate sectoral expenditure across origins according to the same nested homothetic demand system described above. They therefore have the same origin expenditure shares and the same home-foreign elasticity function. If a indexes users and X_{ans} denotes user a 's total expenditure in sector s , then expenditure on origin i is $X_{anis} = \lambda_{nis} X_{ans}$, where λ_{nis} is common across users at every price vector. Summing across users therefore preserves both the origin shares and the home-foreign elasticity function. This assumption concerns sourcing within sectors and does not require users to allocate expenditure across sectors in the same way. This assumption parallels [Caliendo and Parro \(2015\)](#), where the same country-sector composite is used for final consumption and as an intermediate input.

3.2 Nonparametric Identification

Weak separability between home and foreign goods makes each sector's home-foreign elasticity function independent of the price path. Weak separability across sectors allows us to identify this function without specifying the cross-sector aggregator $\mathcal{E}_n$. Without this structure, repeated observations within a country-sector pair could instead identify the path-specific elasticity function characterized in Section 2, provided that the observations trace a given price path and the function is stable over time. In practice, we observe a short time horizon. Hence, we use a single time change for each country-sector pair and assume that the path-invariant elasticity function is common across destinations within each sector. We use cross-sectional variation in domestic expenditure shares and relative-price changes to trace this function. The key cross-country restriction in our baseline is therefore $\varepsilon_{ns}(\lambda) = \varepsilon_s(\lambda)$ for every destination n within sector s . Because we estimate in changes, this restriction permits time-invariant differences in demand levels across destinations.

To connect this restriction to the variation used in estimation, let $\Delta \kappa_{ns}$ denote the change in the foreign-versus-home demand shifter in destination n and sector s . We normalize the foreign aggregate so that this shifter also absorbs the aggregate effect of changes in preferences over foreign sources. By the definition of the home-foreign elasticity, a first-order

expansion around the initial domestic expenditure share gives

$$\Delta \ln \frac{X_{nns}}{X_{nFs}} = \varepsilon_s(\lambda_{nns,0}) \left(\Delta \ln \frac{P_{nFs}}{p_{nns}} - \Delta \kappa_{ns} \right) + \eta_{ns}^H, \quad (13)$$

where η_{ns}^H is the error from applying the local relationship over a finite change; Section 4.4 assesses the local tariff approximation by allowing the elasticity to change over the tariff response while keeping the controls evaluated at the initial share.

To decompose the relative-price change in equation (13), we use a first-order approximation to the foreign price-index change,

$$\Delta \log P_{nFs} = \sum_{i \neq n} \mu_{nis,0} \Delta \log p_{nis} + \eta_{ns}^P,$$

where $\mu_{nis,0}$ is origin i 's initial-period share of country n 's foreign expenditure within sector s , with the shares summing to one across foreign sources. The term η_{ns}^P is the finite-change approximation error from using initial foreign expenditure shares to weight price changes. One benefit of this approach is that it does not require an estimate of the across-foreign-source elasticity. Let $c_{is} = w_i/a_{is}$ denote the unit production cost of country i in sector s , where w_i is the wage in country i and a_{is} is its productivity in sector s , so that

$$\Delta \log c_{is} = \Delta \log w_i - \Delta \log a_{is}.$$

In the model, output prices equal unit production costs. Hence, for each source country i , consumer prices in destination n satisfy

$$\Delta \log p_{nis} = \Delta \log(1 + \mathcal{T}_{nis}) + \Delta \log c_{is} + \Delta \log \mathcal{D}_{nis},$$

where $\mathcal{T}_{nis}$ is the tariff and $\mathcal{D}_{nis}$ is the non-tariff trade cost and where $\mathcal{T}_{nns} = 0$ and $\mathcal{D}_{nns} = 1$. The tariff-induced component of the change in foreign prices relative to home prices is

$$Q_{ns}^{\mathcal{T}} \equiv \sum_{i \neq n} \mu_{nis,0} \Delta \log(1 + \mathcal{T}_{nis}),$$

and the corresponding country-sector unit-cost component of this relative-price change is

$$Q_{ns}^P \equiv \sum_{i \neq n} \mu_{nis,0} (\Delta \log c_{is} - \Delta \log c_{ns}).$$

Define their sum as the component of the change in foreign prices relative to home prices due to tariffs and unit production costs:

$$Q_{ns} \equiv Q_{ns}^{\mathcal{T}} + Q_{ns}^P = \sum_{i \neq n} \mu_{nis,0} [\Delta \log(1 + \mathcal{T}_{nis}) + \Delta \log c_{is} - \Delta \log c_{ns}].$$

Combining the previous expressions gives

$$\Delta \log P_{nFs} - \Delta \log p_{nns} = Q_{ns} + R_{ns},$$

where R_{ns} collects changes in non-tariff trade costs and the foreign-price-index residual:

$$R_{ns} \equiv \sum_{i \neq n} \mu_{nis,0} \Delta \log \mathcal{D}_{nis} + \eta_{ns}^P.$$

Finally, substituting into equation (13) gives the estimating relationship underlying both the nonparametric identification argument and the parametric implementation:

$$\Delta \ln \frac{X_{nns}}{X_{nFs}} = \varepsilon_s(\lambda_{nns,0}) Q_{ns} + u_{ns}, \quad (14)$$

where

$$u_{ns} \equiv \varepsilon_s(\lambda_{nns,0}) (R_{ns} - \Delta \kappa_{ns}) + \eta_{ns}^H.$$

The error term u_{ns} combines changes in non-tariff trade costs and the foreign-price-index residual (collected in R_{ns}); changes in the home-foreign demand shifter (collected in $\Delta \kappa_{ns}$); and the finite-change approximation error η_{ns}^H .

The following proposition gives conditions under which variation in $Q_{ns}^\mathcal{T}$ identifies the elasticity function $\varepsilon_s(\lambda)$.

Proposition 2. *Fix a sector s and let $\mathcal{L}_s \subset (0,1)$ be an interval contained in the support of $\lambda_{nns,0}$ across destinations in that sector. Suppose Q_{ns}^P is observed and all conditional moments below are finite. Let V_{ns} collect predetermined shares and any other controls. Suppose equation (14) holds for a cross-section of destination-sector cells after one time change. For each $\lambda \in \mathcal{L}_s$, suppose that*

$$\text{Var}(Q_{ns}^\mathcal{T} \mid \lambda_{nns,0} = \lambda, Q_{ns}^P, V_{ns}, s) > 0$$

and

$$\text{Cov}(Q_{ns}^\mathcal{T}, u_{ns} \mid \lambda_{nns,0} = \lambda, Q_{ns}^P, V_{ns}, s) = 0.$$

Then the elasticity function is identified on $\mathcal{L}_s$ as

$$\varepsilon_s(\lambda) = \frac{\text{Cov}\left(\Delta \ln \frac{X_{nns}}{X_{nFs}}, Q_{ns}^\mathcal{T} \mid \lambda_{nns,0} = \lambda, Q_{ns}^P, V_{ns}, s\right)}{\text{Var}(Q_{ns}^\mathcal{T} \mid \lambda_{nns,0} = \lambda, Q_{ns}^P, V_{ns}, s)}. \quad (15)$$

Proof. Conditional on $(\lambda_{nns,0}, Q_{ns}^P, V_{ns}, s)$, take the covariance of equation (14) with $Q_{ns}^\mathcal{T}$. Because the conditioning set includes Q_{ns}^P , we have $\text{Cov}(Q_{ns}^\mathcal{T} + Q_{ns}^P, Q_{ns}^\mathcal{T} \mid \cdot) = \text{Var}(Q_{ns}^\mathcal{T} \mid \cdot)$. Orthogonality of $Q_{ns}^\mathcal{T}$ and u_{ns} then gives equation (15). Continuity of ε_s , implied by the maintained demand assumptions and its strict positivity, extends identification from almost every share to all of $\mathcal{L}_s$. $\square$

The tariff component $Q_{ns}^\mathcal{T}$ is an observed component of the relative-price change that enters the regressor in equation (14). The positive conditional tariff variance is the scalar rank condition that allows this variation to identify the elasticity. A sufficient orthogonality condition is that $Q_{ns}^\mathcal{T}$ be conditionally orthogonal to R_{ns} , the demand-shifter change $\Delta \kappa_{ns}$, and the finite-change error η_{ns}^H . This condition permits demand changes and allows them to be correlated with Q_{ns}^P . It requires tariff-induced price variation to be orthogonal to these disturbances conditional on Q_{ns}^P and the other included variables.

We can also identify the elasticity using excluded instruments for both the tariff and measured-price components, while allowing their coefficient functions to differ. This requires the instruments to be orthogonal to the regression error conditional on the initial domestic share and other controls, and to satisfy the corresponding relevance condition. Appendix B.3 states the result. Section 4.4 implements this approach using price changes in other sectors to construct the additional instrument.

Alternatively, we can omit Q_{ns}^P under a different exclusion restriction. The residual then becomes $\varepsilon_s(\lambda_{nns,0})Q_{ns}^P + u_{ns}$. Tariff changes must then be orthogonal to this residual conditional on $(\lambda_{nns,0}, V_{ns}, s)$. With positive conditional tariff variance, the same covariance argument identifies $\varepsilon_s(\lambda)$. Neither this exclusion restriction nor the one in Proposition 2 implies the other. Section 4.4 compares the baseline estimates with specifications that instrument the measured-price terms or omit them.

This nonparametric argument identifies $\varepsilon_s(\lambda)$ only on the range of domestic shares for which there is both support in $\lambda_{nns,0}$ and residual tariff variation within sector s . Calculating gains from trade requires the elasticity function between the observed trade share and one. If that interval extends beyond the support of the identifying variation, the gains calculation necessarily uses functional-form or shape restrictions to extrapolate the elasticity function. As shown in Figure 1 after introducing the data, both empirical samples contain substantial support in initial domestic expenditure shares near one, reducing the extrapolation required along this dimension.

In practice, we approximate the elasticity function using a finite set of coefficients. This restriction allows us to estimate the function with the available data and specifies how we extrapolate beyond the observed domestic expenditure shares. Section 4.1 describes the approximation and its implementation.

4 Estimating Home-Foreign Elasticities

Building on the identification results in Section 3.2, this section estimates sector-specific home-foreign trade elasticity functions. We first describe the parametric implementation, then the data and measurement, and finally the estimation results and a range of sensitivity exercises.

4.1 Parametric Implementation

Parametric Specification. As in Section 3.2, the empirical sample consists of one T -year change for each destination-sector cell. This time change removes the time-invariant component of destination-sector demand heterogeneity, while changes in demand shifters remain in the residual. We use the local estimating relationship in equation (13), where Δ denotes this single time change and $\lambda_{nns,0}$ denotes the initial-period domestic expenditure share. Section 4.4 considers a finite-change tariff response as a sensitivity exercise.

Spline approximations allow us to replace the unknown elasticity function with a finite set of coefficients (Newey, 1997). In practice, we use a restricted cubic spline with three knots (set at three values of the domestic expenditure share) at which the pieces of the function join smoothly. We refer to this as a “restricted” spline because we restrict the function to be

linear below the first knot and above the last knot. The purpose of these linearity restrictions is to reduce erratic behavior near the ends of the observed range (Harrell, 2015). They also simplify our extrapolation when the gains calculation requires shares beyond those observed in the data.

To describe the approximation, choose three domestic expenditure shares $0 \leq a < b < c \leq 1$. These values represent the three knots. Let $(x)_+ = \max\{x, 0\}$ denote the value of x when it is positive and zero otherwise, and define

$$R(\lambda) = \frac{(\lambda - a)_+^3 - \frac{c-a}{c-b}(\lambda - b)_+^3 + \frac{b-a}{c-b}(\lambda - c)_+^3}{(c - a)^2}.$$

We write the elasticity function imposing the same restriction as in section 3.2, that it is common across countries within each sector,

$$\varepsilon_s(\lambda) = \theta_{s0} + \theta_{s1}\lambda + \theta_{s2}R(\lambda).$$

Whereas the function is linear below the first, a , and above the last knot, c , between these knots it is cubic on each side of the middle knot, b , with its value, slope, and second derivative continuous at all three knots.¹⁴

A three-knot spline adds only one coefficient, θ_{s2} , to a linear function. Indeed, setting $\theta_{s2} = 0$ yields the linear case, which we also estimate. Appendix C provides a quantitative Armington gravity model that delivers such a linear elasticity function, strictly nesting the standard constant-elasticity framework. Appendices D and E provide corresponding Krugman and Melitz-Pareto gravity models. Finally, imposing $\theta_{s1} = \theta_{s2} = 0$ yields the standard constant elasticity case.

We allow the intercept of the elasticity function, θ_{s0} , to vary freely across sectors and partially pool the coefficients determining how the elasticity changes with the domestic share by imposing $\theta_{s1} = \theta_{g(s)1}$ and $\theta_{s2} = \theta_{g(s)2}$, where $g(s)$ denotes the sectoral aggregate containing sector s . Thus, sectors within an aggregate share the same slope at any given domestic expenditure share, while their elasticity levels can differ even conditional on a common value of λ . Across specifications, we vary the extent to which sectors are pooled into groups. Specification (1) places all goods-producing sectors in one group. Specification (2) separates manufacturing and non-manufacturing goods. Specification (3) additionally separates consumer and producer manufacturing.

The resulting elasticity function is

$$\varepsilon_s(\lambda) = \theta_{s0} + \theta_{g(s)1}\lambda + \theta_{g(s)2}R_{g(s)}(\lambda). \quad (16)$$

We construct R_g using the formula above, with knots at the 10th, 50th, and 90th percentiles of the initial domestic expenditure share across observations in group g across all countries. We also estimate the linear case, in which case we impose $\theta_{g2} = 0$ for every group.

Estimating Equation. We use origin-sector output prices to measure the component of foreign-relative-to-home delivered-price changes arising from output-price changes. Let $p_{iis,t}$

¹⁴We establish linearity below the first knot, a , and above the last knot, c , in Appendix A.2.

denote the price of origin i 's sector- s good when sold domestically in year t , expressed in a common currency. Using the initial foreign expenditure shares, we construct

$$Q_{ns}^T = \sum_{i \in \mathcal{O}_{ns}} \mu_{nis,0} \Delta \ln(1 + \mathcal{T}_{nis}), \quad (17)$$

$$Q_{ns}^P = \sum_{i \in \mathcal{O}_{ns}} \mu_{nis,0} (\Delta \ln p_{iis} - \Delta \ln p_{nns}), \quad (18)$$

where $\mathcal{O}_{ns}$ contains the foreign origins with matched tariffs and output-price changes. The weights, $\mu_{nis,0}$, sum to one over these origins. Thus, Q_{ns}^P is the initial-share-weighted change in foreign origins' output prices relative to the destination's output price. We include Q_{ns}^P as a control and allow its coefficients to differ from those on Q_{ns}^T . Below, we state the identifying restrictions and explain this choice.

Both shifters capture components of changes in foreign-to-home prices. A destination-sector that lowers its import tariffs experiences a decline in Q_{ns}^T , with a larger decline when the tariff cuts are concentrated in origins accounting for larger shares of its initial foreign expenditure. Similarly, Q_{ns}^P declines when a destination-sector's domestic output price rises relative to its foreign suppliers' own domestic output prices, with a larger decline when those relative price changes are concentrated in origins accounting for larger initial foreign expenditure shares.

The estimating equation is

$$\begin{aligned} \Delta \ln \frac{X_{nns}}{X_{nFs}} = & [\theta_{s0} + \theta_{g1} \lambda_{nns,0} + \theta_{g2} R_g(\lambda_{nns,0})] Q_{ns}^T \\ & + [\gamma_{s0} + \gamma_{g1} \lambda_{nns,0} + \gamma_{g2} R_g(\lambda_{nns,0})] Q_{ns}^P + \mathbf{A}'_{ns} \boldsymbol{\kappa} + u_{ns}, \end{aligned} \quad (19)$$

where we have used the shorthand $g = g(s)$. The vector $\mathbf{A}_{ns}$ contains group indicators and their interactions with the initial domestic share and its spline term. These additive controls allow demand changes to differ in levels and to vary non-linearly with the initial domestic share across groups. The coefficients on the tariff shifter, Q_{ns}^T , and the measured-price shifter, Q_{ns}^P , have sector-specific intercepts and group-specific coefficients on the initial domestic share and its spline term. In the linear specification, we omit R_g from the elasticity and both sets of controls. The residual contains demand changes remaining after these controls, finite-change and approximation errors, and any measurement error.

Tariff Instruments. Observed tariff changes may respond to changes in demand or other sectoral conditions that also affect home relative to foreign expenditure. Controlling for measured output-price changes need not remove this relationship. We therefore instrument the tariff regressors using changes implied by broad trade agreements, which link tariff changes across sectors to common integration commitments. Our identifying assumption, stated formally below, is that the resulting agreement instruments are orthogonal to the remaining regression error after removing their linear relationship with the included controls.

The broad trade agreements we include in our instrument are the 2004 and 2007 European Union enlargements, the EU-Türkiye customs union, the Gulf Cooperation Council, and the West African Economic and Monetary Union. We also include MERCOSUR, NAFTA,

PAFTA, the Common Economic Zone, AFTA, and the bilateral agreements listed in Appendix A.1. We use the tariff changes implied by these agreements and set other bilateral tariff changes to zero in the instrument. Let z_{nis}^T denote the agreement-implied change in the log gross tariff. We aggregate these changes using the same initial foreign expenditure shares as in equation (17):

$$Z_{ns}^T = \sum_{i \in \mathcal{O}_{ns}} \mu_{nis,0} z_{nis}^T.$$

For each origin-destination pair covered by these agreements, we construct z_{nis}^T as the change from the observed 1995 tariff to the agreement-specific terminal tariff described in Appendix A.1. For all other pairs, we set $z_{nis}^T = 0$, regardless of their observed tariff changes.

For each regressor containing Q_{ns}^T in equation (19), we construct its instrument by replacing Q_{ns}^T with Z_{ns}^T . In the first stage, we project each tariff regressor on the full set of instruments and included controls. In the second stage, the fitted sector-specific tariff regressors have coefficients θ_{s0} , while the fitted group-specific tariff interactions with $\lambda_{nns,0}$ and $R_g(\lambda_{nns,0})$ have coefficients θ_{g1} and θ_{g2} , respectively.

The intuition is simple. After accounting for the controls, a larger response of home relative to foreign expenditure to a given instrument-induced tariff change implies a higher home-foreign elasticity. Within a group, systematically larger responses in one sector at a given initial domestic share imply a higher sector-specific intercept θ_{s0} . If responses are larger at higher initial domestic shares, the estimated elasticity function rises with the share. In the linear specification, this relationship is captured by θ_{g1} . With the spline, θ_{g1} and θ_{g2} jointly determine how the elasticity changes with the share, allowing its slope to vary across the domestic-share interval. Identification of these coefficients requires the conditions stated below.

Moment Conditions and Identification. Our identifying restriction concerns tariff changes implied by agreements, after controlling for measured output prices and the other included terms. To state the restriction, let D_{ns} collect the tariff regressors in equation (19), let Z_{ns} collect their agreement instruments, and let W_{ns} collect the controls. Write the regression as

$$\Delta \ln \frac{X_{nns}}{X_{nFs}} = D'_{ns} \boldsymbol{\theta} + W'_{ns} \boldsymbol{\delta} + u_{ns},$$

where $\boldsymbol{\delta}$ contains the control coefficients. Define the part of each excluded instrument remaining after its linear projection on the controls as

$$\tilde{Z}_{ns} = Z_{ns} - \mathbb{E}[Z_{ns} W'_{ns}] \mathbb{E}[W_{ns} W'_{ns}]^{-1} W_{ns}.$$

For the tariff coefficients to be identified, the residualized instruments must satisfy the exclusion restriction

$$\mathbb{E}[\tilde{Z}_{ns} u_{ns}] = 0 \tag{20}$$

and the relevance, or rank condition

$$\text{rank } \mathbb{E}[\tilde{Z}_{ns} D'_{ns}] = \dim(\boldsymbol{\theta}). \tag{21}$$

The first condition requires the remaining agreement variation to be unrelated to remaining demand shifts, errors in measured prices, unmeasured changes in non-tariff trade costs, and approximation errors. The second condition requires the residualized agreement instruments to generate enough variation in the tariff regressors to distinguish all the elasticity coefficients.

A sufficient, but stronger than necessary, condition for the exclusion restriction is $\mathbb{E}[u_{ns} | Z_{ns}, W_{ns}] = 0$. This requires the regression error to be mean zero conditional on the instruments and controls. Another sufficient condition is that the instruments are uncorrelated with both the regression error and the included controls. In this case, removing the instruments' linear relationship with the controls preserves their orthogonality to the error, even if the controls themselves are correlated with the error. Neither sufficient condition implies the other, and the exclusion restriction in (20) is weaker than either: it requires only that the error be orthogonal to the instruments after their linear projection on the controls has been removed.

Under restrictions (20) and (21), the tariff coefficients identify the home-foreign elasticity while the measured-price coefficients remain unrestricted. We choose this specification for two reasons. First, measured output-price changes may be correlated with demand changes. Their fitted coefficients can therefore absorb both the response to prices and demand changes associated with those prices. Second, the published industry price indices used to construct Q_{ns}^P are proxies for the producer prices in the model. Changes in these indices need not track changes in the relevant prices one-for-one. Suppose, for example, that measured price changes equal a constant $v \neq 1$ times true price changes. The coefficient on the measured-price shifter would then equal the elasticity divided by v , even in the absence of correlated demand changes. Allowing separate coefficients accommodates this scaling difference.

An alternative would impose common coefficients on the two components and instrument their sum, $Q_{ns}^T + Q_{ns}^P$. With the scaling difference described above, however, the sum places the wrong relative weight on its two components. Even an arbitrarily strong instrument generally cannot correct this misspecification. Our approach allows the measured-price coefficients to absorb this difference, while maintaining the conditional exclusion restriction in (20). Moreover, in practice instrumenting the sum also produces very weak conditional first stages.

We consider two further specifications to assess the role of measured-price controls. First, we instrument both the tariff and measured-price terms. In this specification, we residualize the instruments with respect to the additive controls alone. The residualized instruments must be uncorrelated with the regression error and provide enough independent variation to identify both sets of coefficients. We construct the additional price instrument using observed price changes in other sectors, as described in Section 4.4. Second, we omit the measured-price terms. This requires the agreement instruments, after accounting for the additive controls, to be uncorrelated with a residual that also includes the omitted price component. These specifications allow us to compare estimates under different exclusion restrictions.

Estimator and Inference. We estimate equation (19) by unweighted two-stage least squares. Once we construct the domestic shares, $R_g(\lambda_{nns,0})$, and the two price shifters, all

regressors are observed and the equation is linear in its coefficients. There is one excluded instrument for each tariff regressor, so the model is exactly identified. Appendix A.2 gives the implementation details.

Our baseline estimation imposes no sign restrictions on the fitted elasticity functions. We use this estimator as the baseline to show what the data call for under the maintained specification before imposing positivity. In the results section, we report the share of observations with non-negative fitted elasticities. In Section 4.4, we re-estimate the model with the elasticity bounded below by 0.1 throughout $[0, 1]$.

We report standard errors clustered by destination country, allowing errors across sectors within a destination to be correlated. Domestic shares, origin weights, published prices, and knot locations are treated as observed. For each group, we also report the smallest destination-clustered Sanderson-Windmeijer conditional first-stage F statistic across its endogenous regressors (Sanderson and Windmeijer, 2016; Windmeijer, 2024). These statistics describe the strength of the excluded variation conditional on the other endogenous regressors.

Reported Functions of Estimates. A spline has no single slope. We therefore evaluate the derivative of the fitted elasticity function at each sector’s median initial domestic expenditure share and then average equally across sectors in a group. Let $\mathcal{S}_g = \{s : g(s) = g\}$ and let $\lambda_{50,s}$ denote that sector-specific median domestic absorption share. The reported slope is

$$\hat{d}_g = \frac{1}{|\mathcal{S}_g|} \sum_{s \in \mathcal{S}_g} \hat{\varepsilon}'_s(\lambda_{50,s}).$$

For the linear specification, this equals $\hat{\theta}_{g1}$. For the spline, it averages $\hat{\theta}_{g1} + \hat{\theta}_{g2} R'_g(\lambda_{50,s})$ across sectors. Because a positive value does not imply that the function rises at every share, we additionally display the full function.

To summarize elasticity levels, let $\lambda_{p,s}$ denote the p th percentile of the initial domestic expenditure share across destinations within sector s . For each group, we report

$$\hat{\varepsilon}_{p,g} = \frac{1}{|\mathcal{S}_g|} \sum_{s \in \mathcal{S}_g} \hat{\varepsilon}_s(\lambda_{p,s}), \quad p \in \{25, 75\}.$$

Each reported elasticity is the average, taken across sectors, of the fitted elasticity evaluated at each sector’s domestic absorption share at percentile p of its distribution. The nonnegative share is the fraction of destination-sector observations in the group with $\hat{\varepsilon}_s(\lambda_{nns,0}) \geq 0$.

4.2 Data and Measurement

This section describes the variables required to estimate equation (19) and how we construct them using two separate data sets. The baseline uses WIOD for 1995 and 2007. We repeat the estimation using OECD ICIO data for the same years. Both data sets provide the expenditure flows and accompanying country-sector output-price data required for our estimation. We combine these data with tariff changes from the Global Tariff Database (Teti, 2024).

Unlike bilateral trade flows, sectoral domestic expenditure is not observed in a single harmonized international source. Instead, WIOD and the OECD ICIO each construct these variables using national supply-use and input-output tables. These are supplemented using additional sources and various assumptions (Timmer et al., 2015; Organisation for Economic Co-operation and Development, 2025). We use WIOD in our baseline exercises because WIOD’s data involve less imputation; this comes at the cost of fewer countries and more aggregated sectors. The OECD ICIO includes more countries and more disaggregated sectors; but this comes at the cost of more imputation when supply-use tables or detailed import-use tables are unavailable. We therefore use it as a complementary replication.

Variables to Be Measured. For each destination, origin, sector, and endpoint year, estimation requires bilateral expenditure and bilateral tariffs. We also require output-price changes by origin and sector. The input-output tables report absorption at basic prices, which exclude import duties. Let $B_{nis,t}$ denote these values. We add tariff payments to foreign purchases, $X_{nis,t} = (1 + \mathcal{T}_{nis,t})B_{nis,t}$ for $i \neq n$, and leave domestic purchases unchanged, $X_{nns,t} = B_{nns,t}$. We construct the dependent variable, initial domestic share, and initial foreign-origin weights from these expenditures. This adjustment adds import duties; other product taxes and distribution margins are not reconstructed. Appendix A.1 describes the treatment of negative absorption and missing tariffs.

WIOD Measurement. Our baseline expenditure data come from the WIOD 2013 release (Timmer et al., 2015; Timmer, 2012). We use 1995 and 2007. For each origin and sector, we sum intermediate and final expenditure in each destination. After incorporating tariffs, expenditure on the destination’s own goods gives $X_{nns,t}$, and total expenditure on foreign goods gives $X_{nFs,t}$. The estimation sample contains 627 destination-sector observations in 40 countries and 16 goods-producing sectors. These comprise two non-manufacturing goods sectors (agriculture, hunting, forestry, and fishing; mining and quarrying) and 14 manufacturing sectors.

We use changes in log industry output-price indices, expressed in a common currency, as proxies for changes in log domestic producer prices, $\Delta \ln p_{iis}$. The price indices come from WIOD’s companion Socio Economic Accounts, July 2014 release (Groningen Growth and Development Centre, 2014). The variable GO_P is an industry gross-output price index in local currency, normalized to 100 in 1995. The official exchange-rate file reports US dollars per unit of local currency. Our measure of $\Delta \ln p_{iis}$ is therefore

$$\ln \frac{\text{GO_P}_{is,2007}}{\text{GO_P}_{is,1995}} + \ln \frac{\text{EXR}_{i,2007}}{\text{EXR}_{i,1995}}.$$

Both endpoint prices are observed and positive for all 40 countries in all 16 goods sectors.

Tariff changes come from the ISIC industry file in the Global Tariff Database. Following Teti’s recommendation to use winsorized measures, we use `tariff95`, the unweighted average bilateral applied tariff winsorized at the 95th percentile (Teti, 2024). When several industries map to one input-output sector, we average tariff levels with equal weight over industries observed at both endpoints before taking the log change. With tariffs reported in percentage

points, we compute

$$\Delta\tau_{nis} = \ln \left(1 + \frac{\text{tariff95}_{nis,2007}}{100} \right) - \ln \left(1 + \frac{\text{tariff95}_{nis,1995}}{100} \right). \quad (22)$$

The origin weights used in the shifters are

$$\mu_{nis,0} = \frac{X_{nis,1995}}{\sum_{j \in \mathcal{O}_{ns}} X_{njs,1995}}, \quad (23)$$

for each $i \in \mathcal{O}_{ns}$. Thus, the shares sum to one over foreign origins with tariffs observed at both endpoints and matched output-price changes, including origins with zero tariff changes.

OECD ICIO Measurement. Our second expenditure data set comes from the 2025 regular release of the OECD ICIO tables ([Organisation for Economic Co-operation and Development, 2025](#)). We use the same 1995 and 2007 endpoints and the same expenditure construction. The sample contains 2,030 observations in 80 destination countries and 26 goods-producing sectors: 19 manufacturing sectors and seven non-manufacturing sectors. Tariffs use the same source and averaging rule, with the OECD sector concordance described in [Appendix A.1](#).

We use changes in log industry output-price indices, expressed in a common currency, as proxies for changes in log domestic producer prices, $\Delta \ln p_{iis}$. We construct these changes using the OECD’s published 2026 release of tables at previous-year prices ([Organisation for Economic Co-operation and Development, 2026](#)). Let $\text{GO}_{is,t}^{\text{CP}}$ denote the published gross-output total in year t at current prices and $\text{GO}_{is,t}^{\text{PYP}}$ denote the published total for that year valued at the previous year’s prices. Both are reported in US dollars. Our measure of $\Delta \ln p_{iis}$ chains the annual price changes:

$$\sum_{t=1996}^{2007} \ln \frac{\text{GO}_{is,t}^{\text{CP}}}{\text{GO}_{is,t}^{\text{PYP}}}.$$

No additional currency conversion is required. OECD constructs these implicit industry prices using national accounts, STAN, KLEMS, and other national sources, with broader-industry prices and value-added deflators used where output deflators are unavailable. Some detailed country-industry price changes therefore rely on OECD imputations.

Treatment of ROW. In both data sets, $X_{nFs,t}$ includes imports from all foreign origins, including the rest of the world (ROW). ROW lacks matched bilateral tariffs and output prices. We retain expenditure on ROW goods and assign these goods the average observed tariff within the same destination and sector, weighting by imports at basic prices in each year. We include this expenditure when constructing the dependent variable and $\lambda_{nns,0}$. We exclude ROW as a destination in estimation and as an origin when constructing the shifters. Across observations in the all-goods baseline, the median share of initial foreign expenditure accounted for by origins entering the shifters is 91.72 percent in WIOD and 98.37 percent in the OECD ICIO. [Appendix A.1](#) gives the imputation and price-coverage details.

The Distribution of $\lambda_{ns,0}$ and Extrapolation. The gains-from-trade calculation evaluates each sector’s elasticity function from every destination-sector’s initial domestic expenditure share to one. The estimating equation identifies the elasticity at each observed initial domestic expenditure share. Assuming that each sector’s elasticity function is common across countries, variation in these shares across countries allows us to estimate the function over a wider range of shares. How much we must extrapolate to calculate gains from trade then depends on how close the range of observed shares comes to one.

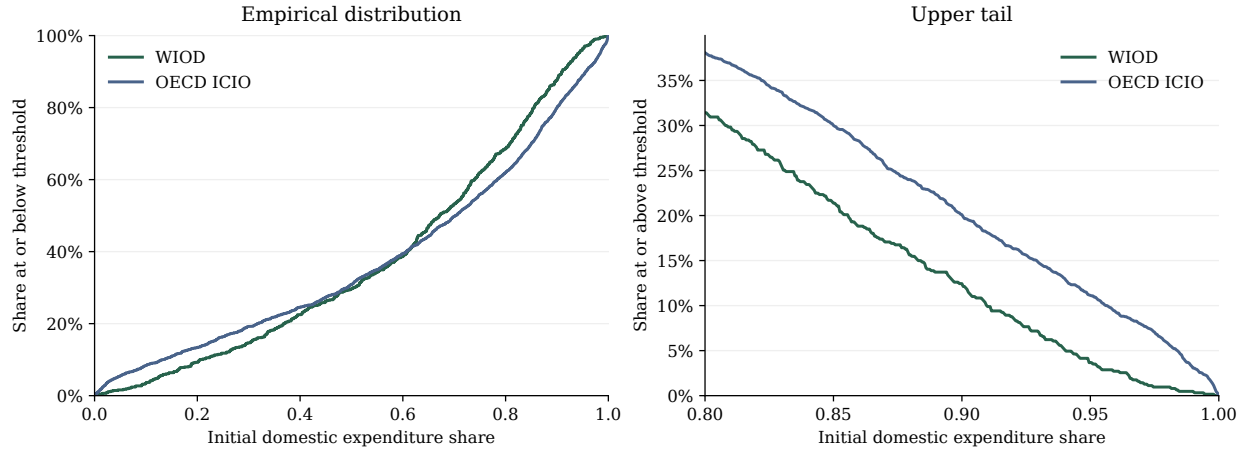


Figure 1: Distribution of Initial Domestic Expenditure Shares in Goods-Producing Sectors

Notes: The figure uses the 1995 domestic shares in the WIOD and OECD ICIO baseline samples. Each destination-sector observation receives equal weight. The left panel reports the share of observations at or below each threshold. The right panel reports the share at or above thresholds of at least 0.8.

Figure 1 displays these distributions. In WIOD and the OECD ICIO, respectively, 12.4 and 20.1 percent of observations have domestic expenditure shares of at least 0.90, while 3.7 and 11.1 percent have shares of at least 0.95. These distributions suggest that the data include observations over much of the range of domestic expenditure shares needed to calculate gains from trade. Two qualifications are important. First, the relevant distribution depends on how sectors are grouped in each specification. Specification (1) pools all goods-producing sectors, so the distributions shown are the relevant ones. Specifications (2) and (3) instead require observations over this range within each of their two or three groups. Second, estimating each group’s elasticity function requires sufficient tariff variation across the range of domestic expenditure shares within that group. Above its last knot, the spline is linear even where we observe domestic expenditure shares. Beyond the highest observed share in a group, calculating gains from trade requires extrapolation. At shares observed in other sectors within the group but not in a particular sector, the calculation relies on the assumption that sectors within the group share the same shape of the elasticity function.

Additional Measurement. To allow the home-foreign elasticity to differ across broad types of manufacturing sectors, we split manufacturing into consumer and producer sectors separately in each data set. For each manufacturing sector, we calculate worldwide final consumption as a share of total absorption in 1995 and classify sectors weakly above the within-data-set median as consumer manufacturing. Final consumption includes household,

nonprofit, and government purchases. Investment and inventories enter total absorption but not final consumption. The classification uses the initial basic-price tables, including ROW, and is held fixed across the estimation exercises. Appendix Tables 5 and 6 report the allocations.

For the country-characteristics sensitivity, PPP GDP per capita and population come from Penn World Table 10.01 (Feenstra et al., 2015), using 1995 values. PPP GDP per capita is expenditure-side real GDP at chained PPPs divided by population. Remoteness is the mean log population-weighted bilateral distance from the destination to all other countries observed in both the Penn World Table and CEPII, weighting each country by its 1995 PPP GDP. Distance and landlocked status come from CEPII GeoDist (Mayer and Zignago, 2011). We standardize log GDP per capita, log population, and remoteness across destination countries and demean landlocked status.

4.3 Estimation Results

Table 1 reports the baseline empirical results, which use WIOD. Panel A uses the three-knot spline and Panel B reports the linear case. Specification (1) uses a single group across all goods, setting $g(s) = g$ for all s . Specification (2) separates manufacturing and non-manufacturing and Specification (3) additionally separates consumer and producer manufacturing. Because the groups are estimated separately (there are group- or sector-specific coefficients), the non-manufacturing estimation results are identical across Specifications (2) and (3).

In Specification (1), the spline’s slope evaluated at sector medians and averaged across sectors is 12.02, with a (destination-clustered) standard error of 4.29. Its 95 percent confidence interval excludes zero. The average fitted elasticity rises from 3.33 at the sector-specific 25th percentiles of the domestic expenditure share to 7.01 at the 75th percentiles. The linear estimates reported in Panel B are very similar. They yield a slope of 12.84, with corresponding elasticity levels of 3.45 and 7.20. Thus, both specifications imply similar qualitative and quantitative results, with higher elasticities at the upper quartile than at the lower quartile.

Separating sectors into distinct groups in Specifications (2) and (3) preserves positive slopes. With the spline, the manufacturing and non-manufacturing slopes are 13.85 and 22.31. Their two-sided p -values are 0.020 and 0.094. Further dividing manufacturing into consumer- and producer-manufacturing in Specification (3) yields slopes of 15.65 and 24.03, respectively; both 95 percent confidence intervals exclude zero. Average fitted manufacturing elasticities are positive at both the 25th and 75th percentiles. For non-manufacturing, the average fitted elasticity is -1.30 at the lower quartile and 5.32 at the upper quartile.

These unrestricted estimates yield negative elasticities at some observed initial domestic expenditure shares $\lambda_{nns,0}$. In Specification (1), 83.3 percent of observations have non-negative fitted elasticities. In Specification (3), this share is 68.0 percent in consumer manufacturing, 87.8 percent in producer manufacturing, and 72.5 percent in non-manufacturing. Section 4.4 reports the effect of imposing a positive lower bound on the full functions (not only across observed initial expenditure shares). The smallest SW F statistic among the spline’s groups is 192.54.

Table 2 repeats the baseline using OECD ICIO expenditures and output prices. The all-goods spline slope at sector medians is 11.72, with a standard error of 5.81, and the average

Table 1: WIOD Estimates of the Home-Foreign Trade Elasticity, 1995–2007

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
<i>A. Three-knot spline</i>							
(1)	All goods	12.02 (4.29)	3.33 (1.22)	7.01 (1.76)	83.3%	4,066.58	627
(2)	Manufacturing	13.85 (5.70)	3.71 (1.33)	7.86 (2.13)	85.0%	2,162.62	547
	Non-manuf. goods	22.31 (13.02)	-1.30 (4.76)	5.32 (4.28)	72.5%	192.54	80
(3)	Consumer manuf.	15.65 (6.64)	1.11 (1.85)	5.32 (2.59)	68.0%	1,144.18	269
	Producer manuf.	24.03 (8.13)	4.76 (1.99)	12.00 (2.69)	87.8%	329.50	278
	Non-manuf. goods	22.31 (13.02)	-1.30 (4.76)	5.32 (4.28)	72.5%	192.54	80
<i>B. Linear schedule</i>							
(1)	All goods	12.84 (4.97)	3.45 (1.23)	7.20 (1.93)	83.6%	3,294.95	627
(2)	Manufacturing	13.88 (5.63)	3.97 (1.42)	8.11 (2.35)	86.7%	1,689.35	547
	Non-manuf. goods	15.21 (11.72)	-0.47 (4.36)	3.34 (3.15)	70.0%	365.73	80
(3)	Consumer manuf.	10.72 (6.13)	2.16 (1.82)	5.78 (2.58)	75.5%	541.47	269
	Producer manuf.	35.43 (11.63)	4.54 (2.09)	13.70 (3.64)	87.8%	480.10	278
	Non-manuf. goods	15.21 (11.72)	-0.47 (4.36)	3.34 (3.15)	70.0%	365.73	80

Notes: “Slope at medians” is the derivative of the fitted elasticity schedule evaluated at each sector’s own median initial domestic expenditure share, averaged across sectors in the row; linear slopes are common throughout the distribution. $\hat{\varepsilon}_p$ evaluates each sector at its own p th percentile of its initial domestic expenditure share and averages across sectors in the row. Share ≥ 0 is the percentage of destination-sector observations with a non-negative fitted elasticity. N is the observation count. Specification (1) pools all goods, (2) separates manufacturing and non-manufacturing, and (3) further separates consumer and producer manufacturing. Destination-clustered standard errors in parentheses. Min. SW F reports the smallest value within the row’s group.

Table 2: OECD ICIO Estimates of the Home-Foreign Trade Elasticity, 1995–2007

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
<i>A. Three-knot spline</i>							
(1)	All goods	11.72 (5.81)	2.14 (1.93)	6.89 (1.75)	72.9%	457.43	2,030
(2)	Manufacturing	9.06 (5.70)	1.07 (2.19)	5.20 (1.70)	68.2%	209.96	1,510
	Non-manuf. goods	43.55 (18.17)	3.40 (2.33)	15.43 (3.16)	80.0%	310.90	520
(3)	Consumer manuf.	13.45 (6.88)	0.43 (2.44)	6.26 (2.04)	74.5%	80.45	793
	Producer manuf.	3.66 (10.29)	2.04 (3.93)	3.93 (2.23)	61.6%	81.95	717
	Non-manuf. goods	43.55 (18.17)	3.40 (2.33)	15.43 (3.16)	80.0%	310.90	520
<i>B. Linear schedule</i>							
(1)	All goods	14.00 (5.25)	2.20 (1.91)	7.07 (1.79)	74.5%	518.55	2,030
(2)	Manufacturing	12.09 (6.22)	1.18 (2.16)	5.33 (1.84)	73.2%	191.25	1,510
	Non-manuf. goods	28.06 (7.72)	5.04 (2.16)	15.14 (3.08)	84.6%	220.73	520
(3)	Consumer manuf.	16.30 (6.15)	0.47 (2.19)	5.98 (2.01)	71.6%	117.22	793
	Producer manuf.	6.42 (14.03)	2.13 (4.19)	4.37 (2.39)	66.9%	57.27	717
	Non-manuf. goods	28.06 (7.72)	5.04 (2.16)	15.14 (3.08)	84.6%	220.73	520

Notes: This table replicates Table 1, using OECD ICIO data.

fitted elasticity rises from 2.14 to 6.89 across the sector-specific quartiles. The consumer-manufacturing slope is 13.45, the producer-manufacturing slope is 3.66 and is imprecisely estimated, and the non-manufacturing slope is 43.55. The share of observations with non-negative fitted elasticities ranges from 61.6% to 80% across groups and the smallest SW F statistic is 80.45. The OECD results therefore also imply positive average slopes. In both data sets, the linear estimates yield the same broad results.

To summarize the levels of the estimated elasticities, we evaluate each sector's fitted function at each observation's initial domestic expenditure share. Under the unrestricted spline in Specification (1), the median across destination-sector observations is 4.65 in WIOD and 2.61 in OECD ICIO. The corresponding interquartile ranges are 1.03 to 8.09 and -0.29 to 6.20. These distributions emphasize that a single constant elasticity would conceal substantial variation.

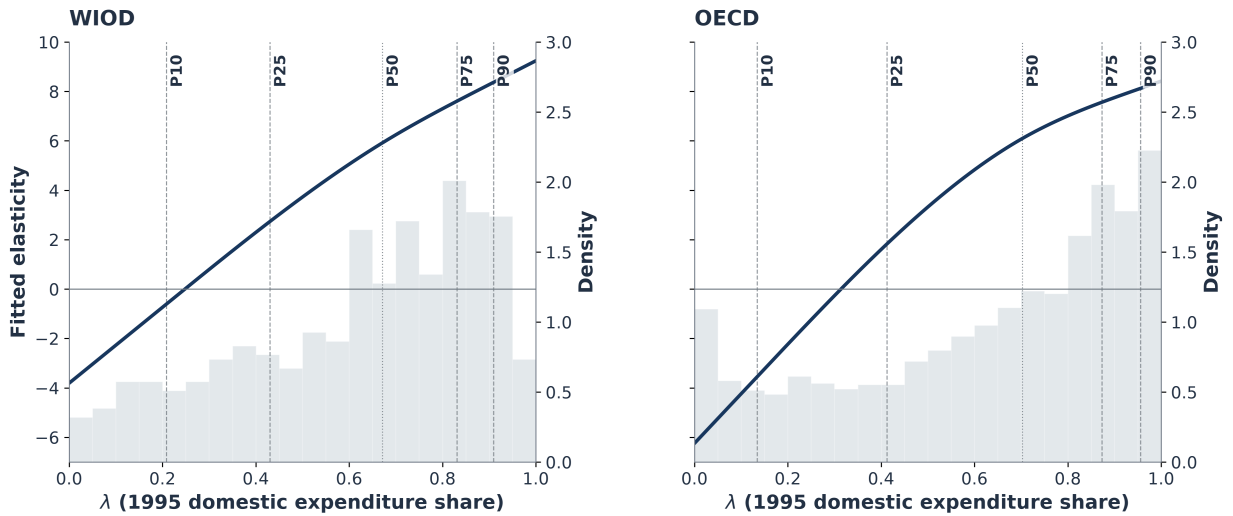


Figure 2: Home-Foreign Elasticity Functions and Domestic Expenditure Shares

Notes: The left panel uses WIOD and the right panel uses OECD ICIO data. Each curve averages fitted elasticities from the unrestricted three-knot Specification (1) equally across sectors at each domestic expenditure share. The histograms show the distributions of observed initial shares across destination-sector pairs on the right axes. Vertical lines mark the 10th, 25th, 50th, 75th, and 90th percentiles within each data set. Both panels use the same elasticity and density scales.

Figure 2 displays the fitted elasticity functions from Specification (1) together with the distribution of initial domestic expenditure shares in each data set. Vertical lines mark the 10th, 25th, 50th, 75th, and 90th percentiles of each distribution. In both WIOD and the OECD ICIO, the fitted elasticity rises throughout the domestic-share interval, although it becomes flatter at higher shares. The functions are negative at sufficiently low shares, but most observations lie at shares where the fitted elasticities are positive. Because the spline knots are located at the 10th, 50th, and 90th percentiles, the fitted functions are linear by construction below the 10th percentile and above the 90th percentile.

4.4 Empirical Sensitivity and Estimator Validation

We consider three broad types of sensitivity exercises using WIOD. First, we examine alternative samples, specifications, and variable constructions. Second, we examine sensitivity to conditioning on measured output prices and use lagged domestic shares to address potentially correlated measurement error. Third, Monte Carlo exercises ask whether the estimator mechanically generates positive slopes when the true functions are flat and whether it recovers positive slopes when the true functions are positive and increasing. We use the spline throughout these exercises, except when changing its functional form explicitly, and in the text we focus attention on results using the all-goods specification, which is Specification (1). The empirical appendix reports all of the corresponding estimates.

Alternative Samples and Specifications. A first set of exercises examines restrictions on variation across countries and sectors as well as sample coverage. First, we allow the elasticity intercept and the measured-price coefficient to vary with the normalized destination characteristics described above. Second, we allow the intercept in $\mathbf{A}'_{ns}\boldsymbol{\kappa}$ to vary by sector, κ_{s0} , rather than group, κ_{g0} . In the first exercise the all-goods average slope at sector medians is 13.07 (reported in Table 7) and in the second (reported in Table 8) it is 14.16, compared with 12.02 in the baseline. All group slopes remain positive in the more disaggregated specifications in both exercises. Third, we examine sensitivity to incomplete tariff coverage. Our baseline constructs the shifters using origins with matched tariff and output-price data, renormalizing their initial foreign expenditure shares to sum to one. To assess whether the results depend on observations for which these origins represent a small share of foreign expenditure, we retain only observations in which they account for at least half of initial foreign expenditure. This reduces the sample from 627 to 591 observations and yields an all-goods average slope at sector medians of 12.68, compared with 12.02 in the baseline (Table 9).

A second set of exercises changes the functional form and examines influential observations. First, we replace the spline with a quadratic functional form, both in the elasticity of interest and the two sets of controls. This yields an all-goods average derivative evaluated at sector medians of 13.83, with consumer- and producer-manufacturing derivatives of 17.31 and 24.25 (reported in Table 10). These are close to the three-knot estimates. Second, we re-estimate each specification after omitting every destination country and every goods sector in turn (reported in Figure 4). The average slopes remain positive in every group and every omission. The all-goods estimate ranges from 8.60 to 14.48 across country omissions and from 8.71 to 14.58 across sector omissions.

A third set of exercises imposes a positive elasticity and assesses the local tariff approximation. First, we require $\varepsilon_s(\lambda) \geq 0.1$ for every sector and every $\lambda \in [0, 1]$.¹⁵ Each slope evaluated at the median remains positive for each grouping in each specification. However, the magnitude of these slopes falls substantially. The all-goods average slope at sector medians falls from 12.02 to 4.81; the consumer, producer, and non-manufacturing slopes fall to 3.82, 12.89, and 6.17 (reported in Table 11). This decline in slopes is intuitive given the

¹⁵Appendix A.2 describes the estimator. We report point estimates only for the constrained exercises; inference would need to account for parameters on or near the boundary (Andrews, 1999; Chernozhukov et al., 2023).

unrestricted estimates. As Figure 2 shows, the fitted elasticity function is negative at very low values of $\lambda_{nns,0}$. Imposing positivity raises the function in this range. To continue fitting the observations at higher domestic shares, the estimated function becomes flatter, reducing its slope. Second, we separately replace the initial-share tariff response with its finite-change counterpart. The all-goods average slope at sector medians is 13.30, with a standard error of 6.04 (reported in Table 12) compared with 12.02 and 4.29 in the baseline local approximation. The group slopes remain positive, although the producer-manufacturing estimate becomes very imprecise. Third, we combine the previous two changes, both imposing the bound $\varepsilon_s(\lambda) \geq 0.1$ for every sector and every $\lambda \in [0, 1]$ and replacing the initial-share tariff response with its finite-change counterpart. The all-goods average slope at sector medians is 4.35, while the consumer-, producer-, and non-manufacturing slopes are 3.28, 7.12, and 6.46 (reported in Table 13). These results are similar to those obtained when only imposing the positivity constraint. The results in Table 13 are those we will match in our quantification, where positivity is required.

Measured-Price Controls. The exclusion restriction in equation (20) requires the agreement instruments to be uncorrelated with the regression error after accounting for the included controls. If measured output prices respond to both agreement-induced tariff changes and unobserved demand changes, removing the instruments' relationship with measured prices can make the remaining instruments correlated with demand changes in the error. We consider two alternatives to assess the role of measured-price controls.

First, we instrument Q_{ns}^P and all its interactions. For each country and sector, we average observed output-price changes in USD equally across the other fifteen goods sectors. We use these averages in place of the sector's own price changes to construct an instrument for Q_{ns}^P . We interact this instrument with the same sector indicators, initial domestic share, and spline term as the measured-price regressors. After accounting for the additive controls, both the agreement instruments and the additional price instruments must be uncorrelated with the regression error. We omit the sector's own price change from the additional instrument to limit correlation arising from sector-specific demand shocks. Table 15 reports the results. The all-goods average slope at sector medians is 10.57, with a standard error of 6.24, compared with 12.02 and 4.29 in the baseline. The corresponding linear estimate is 13.86, compared with 12.84. All spline group slopes remain positive.

Second, we omit Q_{ns}^P and all its interactions, leaving the sample, additive controls, and tariff instruments unchanged. This avoids conditioning on measured prices but requires the agreement instruments to be uncorrelated with a residual that also contains the omitted price component, after accounting for the additive controls. Table 16 reports the results. The all-goods average slope at sector medians is 10.88, with a standard error of 4.64, and the corresponding linear estimate is 13.80. All group slopes remain positive. The similar all-goods estimates across the baseline and these two alternatives are reassuring given their distinct exclusion restrictions.

Correlated Measurement Error. The initial domestic expenditure share, $\lambda_{nns,0}$, is a key covariate, while the initial home-foreign expenditure ratio enters the dependent variable.

Both are constructed from the same expenditure data, since

$$\ln(X_{nns,0}/X_{nFs,0}) = \ln[\lambda_{nns,0}/(1 - \lambda_{nns,0})],$$

where the initial ratio enters the dependent variable with a negative sign. Hence, measurement error that (e.g.) overstates initial domestic expenditure relative to foreign expenditure raises the measured domestic share but lowers the measured change in the home-foreign expenditure ratio. This mechanical relationship can bias the estimated elasticity function.

We address this concern as follows. We re-estimate the model using changes between 1996 and 2007, retaining the 1995 foreign-origin weights, $\mu_{nis,0}$, and the original 1995-2007 agreement shifter, Z_{ns}^T . By estimating starting in 1996, we remove the 1995 home-foreign expenditure ratio from the dependent variable. This allows us to use the 1995 domestic share as an instrument for its 1996 value.

We proceed in two steps. First, in Table 17 we present estimates of the baseline specification using 1996-2007 changes and 1996 domestic shares, without instrumenting the 1996 shares with their 1995 values. This specification shows whether the results change simply because the initial year changes. Second, in Table 18 we present results in which we instrument every regressor constructed using $\lambda_{nns,1996}$ with its counterpart constructed using $\lambda_{nns,1995}$. Appendix A.2 gives the full construction.

The all-goods average slope at sector medians is 11.09 in the current-share specification and 11.13 in the lagged-share specification, with standard errors of 3.26 and 3.53. The other group slopes remain positive. The smallest conditional first-stage statistic in the lagged-share specification is 35.40. The similarity of these estimates is reassuring about correlated measurement error. This interpretation requires that measurement error in $\lambda_{nns,t}$ be uncorrelated between 1995 and 1996. Even under this condition, the exercise need not eliminate measurement-error bias in the non-linear share terms.

Monte Carlo Exercises. Appendix H reports Monte Carlo exercises designed to answer two questions. First, does our estimation procedure mechanically generate positive elasticity-function slopes when the true slopes are zero? Second, does it accurately recover positive slopes when the data are generated by a model with positively-sloped linear home-foreign elasticity functions? We use the quantitative multi-sector Armington model in Appendix C. The simulated economy matches the WIOD baseline’s 40 countries and 16 goods sectors. Four calibrations share the same initial equilibrium but have constant elasticities, a common slope of four, separate manufacturing and non-manufacturing slopes, or separate consumer-manufacturing, producer-manufacturing, and non-manufacturing slopes. We use the same 200 independently drawn tariff-shock matrices in each calibration and solve for the resulting equilibria.

We construct the measured-price control from equilibrium unit-cost changes and estimate the spline specification using the randomized tariffs as their own instruments. The measured-price-control intercept varies by sector, while the additive-control intercept varies by group. Under constant elasticities, the mean estimated all-goods slope evaluated at sector medians is 0.002. With a true common slope of four, it is 4.03. In the two-group calibration, the mean estimates are 3.02 for manufacturing and 6.07 for non-manufacturing, compared with true slopes of three and six. In the three-group calibration, the mean estimates are 3.02, 5.04,

and 7.08 for consumer manufacturing, producer manufacturing, and non-manufacturing, compared with true slopes of three, five, and seven.

5 Quantification

In this section we quantify gains from trade relative to autarky, building on equation (11), but now allowing output from all sectors to be used as intermediate inputs. We take equilibrium 0 to be the 2007 trade equilibrium when using either the WIOD or OECD ICIO data and estimates. Equilibrium 1 is autarky, so that $\lambda_{nns}^1 = 1$ in every traded sector. We therefore raise each sector’s domestic expenditure share from its observed value in the corresponding trade equilibrium to one and calculate the resulting increase in aggregate unit expenditure. This increase determines the ratio of real income in the initial trade equilibrium to real income under autarky. The calculation uses observed tariff-inclusive expenditure shares and assumes that income changes in proportion to factor earnings, abstracting from tariff revenue and international transfers. We focus on the WIOD quantification in the body and use the corresponding OECD ICIO exercise as robustness.

According to the theory in Section 3.1, the increase in each traded sector’s unit expenditure function as its domestic expenditure share rises to one is determined by its initial domestic expenditure share and its home-foreign elasticity function. As also shown there, translating these sectoral changes into aggregate welfare requires the demand system that aggregates consumption across sectors. Section 3.1 allows us to estimate the elasticity functions from total absorption by imposing common sourcing across intermediate and final users within each sector, but it leaves their roles in production and welfare unspecified. For quantification, intermediate-input use determines how sectoral cost changes feed into production costs, while the treatment of final uses connects total absorption to household welfare. We next specify this additional structure.

We retain 16 goods-producing sectors from WIOD: 14 manufacturing sectors and two non-manufacturing goods sectors. Let $\mathcal{S}_M$ and $\mathcal{S}_G$ denote these two sets. We combine the remaining sectors into a single composite, indexed by 0, which we treat as non-traded. We use estimates from Specifications (1), (2), and (3) of Appendix Table 13, which impose $\varepsilon_s(\lambda) \geq 0.1$ for every sector and every $\lambda \in [0, 1]$ and use the finite-change tariff response. Our primary exercise uses Specification (3): the full fitted sector-specific elasticity functions have sector-specific intercepts and separate spline shapes for consumer manufacturing, producer manufacturing, and non-manufacturing goods. We use 2007 expenditure shares for all 640 WIOD country-sector cells. Thirteen cells are excluded from estimation because domestic expenditure is zero in either 1995 or 2007, making the change in the log home-foreign expenditure ratio undefined. For these cells, we use the elasticity function estimated for the corresponding sector. We interpret the welfare calculation at a zero domestic expenditure share as the limit as that share approaches zero.

5.1 Intermediate Inputs and Final Uses

The common-sourcing assumption in Section 3.1 concerns substitution within sectors: final and intermediate users purchase the same sectoral composite and therefore have the same ori-

gin shares and home-foreign elasticity. It leaves substitution across sectors unrestricted. For quantification, we additionally assume that all users purchase the same aggregate composite of sectoral goods and that every production sector has the same value-added share. These additional restrictions are not required for estimating the home-foreign elasticity functions.

In country n , every production sector has the same value-added share $\beta_n \in (0, 1)$ and uses as its intermediate input the same aggregate composite used for household consumption, government purchases, and investment. Let $P_n \equiv \mathcal{E}_n((e_{ns})_{s \in \mathcal{S}})$ denote the price index for this composite. We assume that the unit cost in sector s is

$$c_{ns} = a_{ns}^{-1} w_n^{\beta_n} P_n^{1-\beta_n}.$$

This deliberately parsimonious structure yields the familiar scalar intermediate-input amplification of gains from trade (Arkolakis et al., 2012). Unlike the common-sourcing assumption within sectors, the restrictions on cross-sector expenditure weights and value-added shares are stronger than those in Caliendo and Parro (2015), who allow both to differ across production sectors. We impose these restrictions to keep the quantification focused on the role of home-foreign elasticity functions rather than heterogeneity in production networks.

It would be possible to extend the quantification to allow users to differ in their expenditure allocations across sectors and producers to differ in their value-added shares while retaining the estimated home-foreign elasticity functions. Relaxing common sourcing within sectors would require revisiting the estimation itself, because expenditure aggregated across users would no longer generally identify a common home-foreign elasticity function.

Let γ_{nC} , γ_{nG} , and γ_{nI} denote the fixed shares of nominal income allocated to household consumption, government purchases, and investment, with $\gamma_{nC} + \gamma_{nG} + \gamma_{nI} = 1$. We treat government purchases and investment as current final expenditure that does not enter household utility directly, although it can affect welfare through equilibrium income and prices. We do not model capital accumulation. Household real consumption then equals $\gamma_{nC} Y_n / P_n$. Because γ_{nC} is the same in trade and autarky, the proportional change in household real consumption equals the proportional change in Y_n / P_n .

We calibrate β_n as economy-wide value added divided by gross output in the relevant trade equilibrium. For the WIOD exercise, this is the sum of the value-added row divided by the sum of the output row across country n 's production sectors in 2007. For OECD ICIO, it is the sum of the VA row divided by the sum of the OUT row in 2007. The median β_n is 0.47 in the WIOD quantification sample and 0.48 in the OECD ICIO sample.

5.2 Preferences

We use nested CES preferences across sectors. At the lower tier, manufacturing sectors are aggregated into a manufacturing composite and non-manufacturing goods sectors are aggregated into a non-manufacturing-goods composite. For $b \in \{M, G\}$, let E_{nb} denote the exact price index (unit expenditure function) for composite b ,

$$E_{nb}((e_{ns})_{s \in \mathcal{S}_b}) = \left[\sum_{s \in \mathcal{S}_b} \omega_{nbs} e_{ns}^{1-\sigma_L} \right]^{1/(1-\sigma_L)},$$

where σ_L is the elasticity of substitution across sectors within each goods-producing composite, assumed common across the manufacturing and non-manufacturing-goods composites. At the upper tier, the unit expenditure function aggregates the manufacturing composite, the non-manufacturing-goods composite, and the non-goods composite:

$$\mathcal{E}_n(E_{nM}, E_{nG}, e_{n0}) = [\omega_{nM} E_{nM}^{1-\sigma_U} + \omega_{nG} E_{nG}^{1-\sigma_U} + \omega_{n0} e_{n0}^{1-\sigma_U}]^{1/(1-\sigma_U)},$$

where σ_U is the elasticity of substitution across the three upper-tier composites. Our baseline imposes $\sigma_L > \sigma_U$. The preference weights are country specific and calibrated to reproduce each country's 2007 expenditure shares.

For each goods-producing sector, define its unit-expenditure increase moving to autarky, relative to its domestic price, by

$$\widehat{e}_{ns} \equiv \exp \left(\int_{\lambda_{nns}^T}^1 \frac{d \ln \lambda}{\varepsilon_{ns}(\lambda)} \right).$$

Under the spline elasticity estimated in our baseline specification, we evaluate this integral numerically using

$$\varepsilon_s(\lambda) = \widehat{\theta}_{s0} + \widehat{\theta}_{g(s)1} \lambda + \widehat{\theta}_{g(s)2} R_{g(s)}(\lambda).$$

By construction, $\widehat{e}_{n0} = 1$ for the non-traded non-goods composite. Let $\alpha_{ns|b}$ denote sector s 's initial expenditure share within composite b and let α_{nM} , α_{nG} , and α_{n0} denote the initial upper-tier expenditure shares. The exact CES changes in the two lower-tier price indexes are

$$\widehat{E}_{nb} = \left[\sum_{s \in \mathcal{S}_b} \alpha_{ns|b} \widehat{e}_{ns}^{1-\sigma_L} \right]^{1/(1-\sigma_L)}, \quad b \in \{M, G\}.$$

Under the production structure in Section 5.1, the resulting gains from trade are

$$\ln \frac{W_n^T}{W_n^A} = \frac{1}{\beta_n(1-\sigma_U)} \ln \left[\alpha_{nM} \widehat{E}_{nM}^{1-\sigma_U} + \alpha_{nG} \widehat{E}_{nG}^{1-\sigma_U} + \alpha_{n0} \right]. \quad (24)$$

Thus, initial domestic expenditure shares and the home-foreign elasticity functions determine the sectoral costs of autarky, $\widehat{e}_{ns}$. The lower-tier elasticity σ_L aggregates these sectoral costs into the composite costs $\widehat{E}_{nb}$, while σ_U aggregates the composite costs and $1/\beta_n$ captures their amplification through intermediate inputs. Appendix F derives equation (24).

We choose σ_U using estimates from the structural-change literature, in which substitution across broad sectors is a central concern. Our baseline sets $\sigma_U = 0.6$. Comin et al. (2021) estimate an elasticity of 0.57 using a country-level panel of 39 countries with sectoral employment, value added, and prices together with consumption per capita. Cravino and Sotelo (2019) obtain a final-consumption estimate of 0.59 from U.S. input-output use tables for 1977-2012, using final-consumption shares at producer prices and sectoral gross-output price indexes. Our baseline sets $\sigma_L = 2$. Berlingieri et al. (2026, Appendix Table D.1, column 3) estimate a between-sector elasticity of 2.03 when French sectors are defined at the two-digit level, an aggregation roughly comparable to the WIOD manufacturing sectors used here. In robustness we set $\sigma_L = 1$ and $\sigma_L = 3$. The former imposes Cobb-Douglas aggregation within each goods-producing composite, while the latter provides a higher-substitution comparison.

5.3 Quantitative Results

Table 3: Gains from Trade Across Elasticity-function Specifications

	p10	p25	Median	Mean	p75	p90
(1)	5.75	10.21	15.50	21.00	30.66	39.04
(2)	5.50	9.95	16.13	27.48	31.72	50.00
(3)	7.68	12.75	19.37	32.00	35.17	58.87

Notes: Entries report $100 \ln(W_n^T/W_n^A)$ calculated using equation (24), where β_n is 2007 economy-wide value added divided by gross output. The percentiles and mean are calculated across the 40 countries in the WIOD quantification sample. All three specifications use the finite-change, positive-floor spline estimates in Table 13 and set $\sigma_L = 2$ and $\sigma_U = 0.6$.

Table 3 reports the distribution of gains from trade across the 40 WIOD countries using elasticity functions estimated in Specifications (1), (2), and (3). Under Specification (3), the median and mean gains from trade are 19.37 and 32.00 log points, while the 10th and 90th percentiles are 7.68 and 58.87 log points. Thus, the estimated gains are substantial for the typical country and vary considerably across countries. While results vary across the three specifications, we use Specification (3) in what follows because it is the most disaggregated.

Cross-country variation in gains from trade is positively related to the initial aggregate import share. Panel A of Figure 3 plots this relationship under Specification (3). The import share is measured as expenditure on imports in the 16 goods-producing sectors divided by total expenditure, including expenditure on the outside composite. Its correlation with gains from trade is 0.42. This relationship is intuitive: the aggregate import share measures how far the initial trade equilibrium is from autarky. The relationship is not one-to-one, however. The same aggregate import share can arise from different combinations of sectoral expenditure shares and within-sector domestic expenditure shares. Because these objects interact with sector-specific elasticity functions, countries with similar aggregate import shares can have different gains from trade. The two largest deviations in Panel A of Figure 3 illustrate this composition effect: Luxembourg (LUX) and Belgium (BEL) have low domestic expenditure shares in non-manufacturing goods sectors whose fitted elasticities are low. In addition, countries differ in their value-added shares. This is a separate source of variation: holding the other objects fixed, a lower β_n raises measured gains through the intermediate-input multiplier $1/\beta_n$.

To assess the quantitative implications of allowing the home-foreign elasticity to vary along the path to autarky, we consider a fixed-elasticity comparison. While there is no unique way to construct such a comparison, in Panel B of Figure 3, we hold each country-sector elasticity fixed at its fitted 2007 value, $\varepsilon_{ns,2007} \equiv \widehat{\varepsilon}_s(\lambda_{nns}^{2007})$. This comparison isolates the implications of allowing the elasticity to vary after the initial equilibrium. The comparison retains Specification (3) and the baseline values of the cross-sector elasticities. Moreover, it uses home-foreign elasticities estimated directly from variation in home relative to foreign expenditure. Thus, even the fixed-elasticity calculation differs from the standard approach of using elasticities estimated from substitution across foreign sources. Panel B of Figure 3 plots the gains from trade under the fixed-elasticity calculation against the baseline estimates using the variable elasticity functions. The dashed line is a 45-degree line.

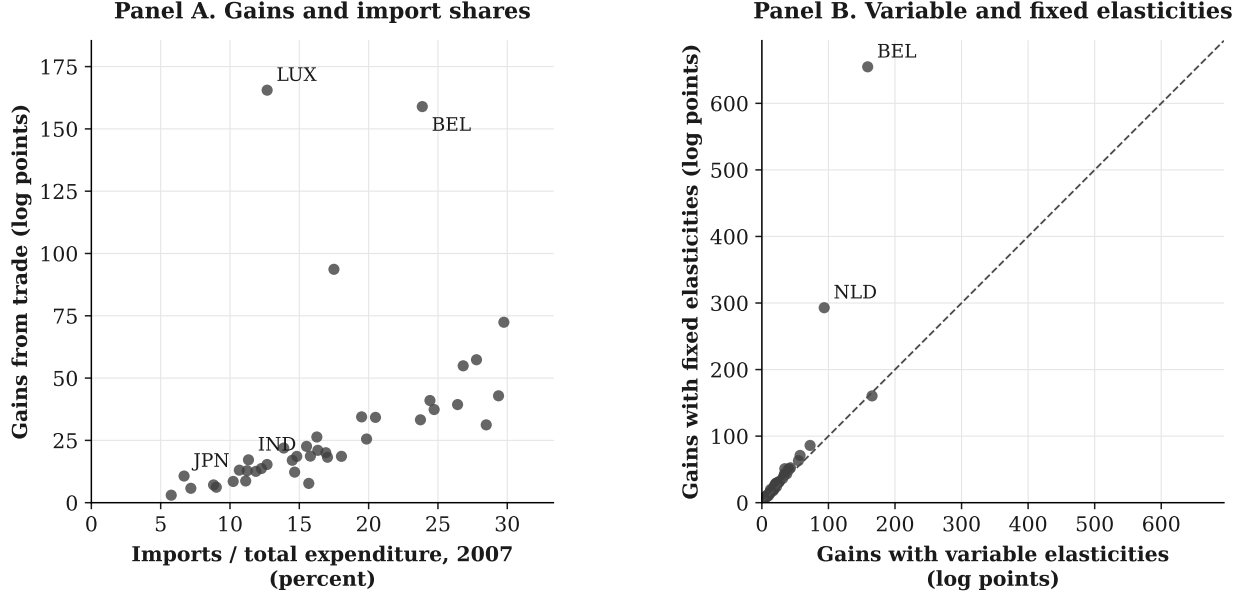


Figure 3: Gains from Trade, Import Exposure, and Variable Elasticities

Notes: Each observation is one of the 40 countries in the WIOD quantification sample. Both panels use estimates from Specification (3) in Table 13, with $\sigma_L = 2$, $\sigma_U = 0.6$, and the country-specific intermediate-input multiplier $1/\beta_n$. Gains from trade are measured as $100 \ln(W_n^T/W_n^A)$. Panel A plots gains against 2007 expenditure on imports in the 16 goods-producing sectors divided by total expenditure, including the non-traded composite. Panel B plots gains holding each country-sector elasticity fixed at its fitted 2007 value against gains under the variable-elasticity functions. The dashed line is a 45-degree line. Selected countries are labeled.

When each sector's elasticity function is increasing throughout the path to autarky, holding its elasticity fixed at its 2007 value produces larger gains from trade. The intuition for this result is simple. When the baseline elasticity rises along the path to autarky, the fixed elasticity lies below the baseline elasticity at every point along the path except for the original point, where they are equal. This implies larger gains from trade under the fixed-elasticity calculation. In our quantification, the variable- and fixed-elasticity measures are closely related, but the difference in magnitudes is substantial: fixing elasticities at their 2007 values raises the median gains from 19.37 to 24.43 log points and the mean from 32.00 to 54.42 log points. The estimated functions used for quantification, which impose a lower bound of 0.1 on elasticities, are not everywhere increasing, however. In particular, the fitted consumer-manufacturing and non-manufacturing functions decrease at low domestic expenditure shares, so the ordering need not hold for every country. Nevertheless, holding elasticities fixed increases gains from trade for 39 of the 40 countries, with Luxembourg the exception.

The two outliers in Panel B, Belgium and the Netherlands, illustrate how fixing low initial elasticities can produce very large gains. In agriculture, forestry, and fishing, their fitted initial elasticities are approximately 0.11 and 0.10, respectively. Holding these elasticities fixed throughout the path to autarky produces much larger sectoral price increases than allowing them to vary. Limited substitution between the non-manufacturing goods composite and

the other composites amplifies the effect on aggregate gains. Allowing non-manufacturing elasticities to vary while keeping manufacturing elasticities fixed yields gains close to the baseline.

These calculations are related to the between-sector mechanism emphasized by [Ossa \(2015\)](#): cross-sector heterogeneity in constant trade elasticities can magnify gains from trade. Under Cobb-Douglas aggregation, a sector’s contribution to gains is the product of its trade exposure, $\alpha_{ns}[-\ln \lambda_{nns}^T]$, and its inverse trade elasticity. Trade exposure is approximately $\alpha_{ns}(1-\lambda_{nns}^T)$, which is the share of country n ’s aggregate expenditure that is spent on imports in sector s . Holding the cross-sector distributions of these two objects fixed, increasing the covariance between trade exposure and inverse trade elasticities further magnifies gains from trade. Our framework nests this calculation when cross-sector aggregation is Cobb-Douglas and the home-foreign elasticity in each sector is constant, and therefore incorporates the between-sector mechanism. [Figure 3](#), however, does not isolate its quantitative importance: both axes retain differences in elasticity levels across sectors. Instead, the figure compares the variable-elasticity functions with home-foreign elasticities held fixed at their initial values along the path to autarky.

Our framework also introduces a distinct within-sector mechanism, which is most directly isolated in [Figure 3](#). When the home-foreign elasticity rises with the domestic expenditure share, a sector with a lower initial domestic expenditure share, for a given elasticity function, begins at a lower elasticity and travels along a longer portion of the elasticity function on the path to autarky; consequently, a larger portion of its gains-from-trade integral is evaluated at low elasticities. For a given sectoral expenditure share, its initial domestic expenditure share therefore determines both its trade exposure and the portion of the elasticity function relevant for its contribution to gains from trade. The connection to the between-sector mechanism is that both give greater welfare weight to low elasticities in sectors with greater trade exposure. The within-sector mechanism, however, operates even if every sector has the same home-foreign elasticity function.

Table 4: Gains from Trade Sensitivity to Cross-Sector Substitution

σ_L	σ_U	p10	p25	Median	Mean	p75	p90
2	0.6	7.68	12.75	19.37	32.00	35.17	58.87
1	0.6	9.85	40.03	76.12	∞	521.98	∞
3	0.6	6.34	9.89	14.77	24.33	25.42	41.70
2	1	7.58	12.05	18.39	25.49	32.41	54.45

Notes: Entries report $100 \ln(W_n^T/W_n^A)$ calculated using equation (24), where β_n is 2007 economy-wide value added divided by gross output. The percentiles and mean are calculated across the 40 countries in the WIOD quantification sample. Specification (3) in [Table 13](#) is used throughout. The baseline sets $\sigma_L = 2$ and $\sigma_U = 0.6$; each subsequent row changes the elasticities shown.

Sensitivity. Finally, [Table 4](#) reports robustness of the [Specification \(3\)](#) results to alternative values of the two cross-sector elasticities. The first row replicates the third row of [Table 3](#). The next two rows lower and raise σ_L from 2 to 1 and 3. Raising σ_L mitigates the impact of a move to autarky because countries can more easily substitute consumption

away from sectors in which price indices rise the most. When $\sigma_L = 1$, nine countries have infinite gains because WIOD records zero domestic expenditure in at least one traded sector and Cobb-Douglas preferences make every sector essential. The median among all countries remains finite at 76.12 log points, but the mean is infinite. Raising σ_L to 3 lowers the median to 14.77 log points. The final row raises σ_U from 0.6 to one while holding $\sigma_L = 2$. The higher upper-tier elasticity allows consumption to shift more easily toward the outside composite as the prices of the two goods-producing composites rise; the median falls more modestly, from 19.37 to 18.39 log points. Appendix Table 19 shows that lowering the elasticity bound from 0.1 to 0.05 leaves median gains nearly unchanged but raises mean gains from 32.00 to 40.55 log points.

In Appendix G, we repeat the quantification using the OECD ICIO data and the corresponding constrained finite-change estimates in Appendix Table 14, taking the 2007 trade equilibrium as the initial equilibrium. Under Specification (3), the mean gains from trade are 26.90 log points, compared with 32.00 using the WIOD data and estimates.

6 Conclusion

Two broad messages emerge from our analysis. The first is negative. The elasticity relevant for gains from trade describes how expenditure on all foreign goods changes relative to expenditure on home output as foreign prices rise relative to the home price. In practice, this elasticity is typically estimated from the relationship between expenditure across foreign countries and their relative prices. Using such an estimate to calculate gains from trade either incorrectly measures gains or requires strong assumptions equating very different margins of substitution. Moreover, typical approaches additionally impose a constant elasticity along the path to autarky. These assumptions may or may not be appropriate in particular contexts, but they restrict the demand system beyond homotheticity and are not required for quantifying the gains from trade.

The second message is more positive. For measuring gains from trade, a more direct empirical approach exists: estimate how home expenditure changes relative to aggregate foreign expenditure as foreign prices change relative to the home price. The logic parallels Arkolakis et al. (2012). They argue that leveraging the relationship between bilateral imports and bilateral trade costs is attractive for welfare analysis because it measures the welfare-relevant response of trade to trade costs rather than recovering that response indirectly from a micro-level parameter. We apply the same argument, but take it a step further, directly estimating the elasticity that enters the gains-from-trade calculation rather than inferring it from substitution across foreign sources. This direct approach is possible because our theoretical results express gains from trade in terms of the initial domestic expenditure share and the home-foreign elasticity function along a price path to autarky. Making the approach empirically tractable requires additional structure: under general homothetic preferences, responses observed along different price paths need not identify the same elasticity function. We therefore impose weak separability between home and foreign goods within each sector, which makes that sector's elasticity function independent of the price path. Together with our assumption that the function is common across destinations within a sector, this allows us to combine price variation across countries to estimate the function over the supported

range of domestic expenditure shares.

For future structural applications, we develop Armington, Krugman, and Melitz-Pareto gravity models in which the home-foreign elasticity is exactly linear in the domestic expenditure share rather than constant, while substitution across foreign sources is governed by a separate elasticity. These models retain the tractability of familiar gravity frameworks while permitting variable home-foreign elasticity functions in quantitative counterfactual analysis. Because they specify the full general equilibrium, they can also be used to solve for the effects of counterfactual changes in trade costs and other primitives. The movement to autarky studied here is only one such counterfactual.

In the empirical analysis, we estimate the welfare-relevant elasticity function directly from changes in home relative to foreign expenditure. Across two independently constructed global data sets, WIOD and the OECD ICIO, the estimated home-foreign elasticities generally rise with domestic expenditure shares. The average slope at sector medians in the all-goods specification is positive and statistically significant in both data sets. The broad pattern is stable across alternative samples, sectoral classifications, variable constructions, and instrumental-variable strategies.

The paper introduces a novel within-sector connection between trade exposure and the elasticity relevant for welfare. For a given increasing elasticity function, a destination-sector pair with a lower initial domestic expenditure share begins the path to autarky at a lower home-foreign elasticity. Hence, a larger portion of its gains-from-trade integral is evaluated at low elasticities. Quantitatively, using the positive finite-change WIOD estimates, the mean gain from trade is 32.00 log points when the elasticity is allowed to vary along the path to autarky, compared with 54.42 log points when each country-sector elasticity is held fixed at its initial value.

The analysis also indicates where further evidence would be most valuable. Sectoral domestic expenditure is constructed less directly than international trade flows, although the similarity of the findings across the WIOD and OECD ICIO data is reassuring. In addition, the calculation of gains from trade requires evaluating elasticity functions all the way to autarky. Both data sets include a substantial share of observations with high domestic expenditure shares; nevertheless, the exercise relies on the functional form of the elasticity function and on restrictions that allow observations from different countries and, in some specifications, sectors to inform one another. Better measurement of domestic expenditure and additional price variation over a wider range of domestic expenditure shares would permit these restrictions to be relaxed. Longer panels could also permit estimation of path-specific home-foreign elasticity functions without imposing weak separability, provided that observations within each country-sector pair trace a given price path and the function is stable over time.

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Empirical Appendix

A Additional Empirical Details

A.1 Additional Data Details

Country Characteristics. For the WIOD sensitivity exercises, log PPP GDP per capita, log population, and remoteness use 1995 values. We subtract the mean across destination countries and divide by the sample standard deviation. We demean the landlocked indicator. For remoteness, we match the Democratic Republic of the Congo, Palestine, and Romania to their older CEPII codes. We use the CEPII distances for Yugoslavia for Serbia and Montenegro, weighting them by their combined 1995 PPP GDP. The exercise adds these characteristics to the elasticity intercept and the coefficient on measured-price changes, with separate coefficients that are common across groups.

WIOD Measurement. The WIOD 2013 world input-output tables cover 40 named countries and ROW from 1995 to 2011 ([Timmer et al., 2015](#); [Timmer, 2012](#)). For each destination, we sum intermediate and final uses to construct what we refer to as “basic-price absorption,” $B_{nis,t}$. Inventory changes are sometimes negative and can make the sum of intermediate and final uses negative. When bilateral absorption is negative, we set it to zero. We retain destination-sector observations with positive domestic and foreign expenditure in both years and an observed change in the domestic output price. The manufacturing subset contains 547 observations in 14 sectors. Adding agriculture/hunting/forestry/fishing and mining/quarrying gives the 627-observation baseline.

The companion Socio Economic Accounts report the gross-output price index GO_P , with 1995 equal to 100 ([Groningen Growth and Development Centre, 2014](#)). Table 7.1 of [Timmer \(2012\)](#) describes this variable. We use the official exchange-rate series to convert its change to US dollars. The price index and exchange rates use the currency conventions of the same WIOD release. This gives observed country-sector price changes for every named origin in all sixteen goods sectors.

OECD ICIO Measurement. For each origin-sector row in the ICIO table, we sum absorption across every intermediate-use column and the destination’s final-demand columns: HFCE, NPISH, GGFC, GFCF, INVNT, and DPABR. We treat negative inventories and negative aggregate bilateral absorption as in WIOD. The current-price tables and previous-year-price tables share the 80-country and 50-industry classification ([Organisation for Economic Co-operation and Development, 2025, 2026](#)). Annual output margins allow us to construct price changes directly in each sector without mapping WIOD prices to OECD sectors.

Valid price changes are available for 1,518 of the 1,520 named-country manufacturing cells. Brunei produces no motor vehicles and Luxembourg produces no refined petroleum in any of the years compared. Their missing prices therefore do not affect origins with positive initial expenditure weights. In non-manufacturing, Germany has some positive sales of metal ores, but its price change is unavailable. We include these sales in total foreign expenditure but exclude them when constructing the price and tariff shifters. The final sample contains

1,510 manufacturing observations and 520 observations in agriculture, forestry, fishing, coal, oil and gas, metal ores, and other mining.

Tariff-Inclusive Expenditure and ROW. The published input-output flows are at basic prices. We multiply each observed foreign flow by its endpoint gross tariff before constructing the dependent variable, domestic share, and initial foreign-origin shares. Domestic flows are unchanged. For ROW and other foreign flows without an observed tariff, we use the destination-sector average observed tariff, weighted by basic-price foreign expenditure at that endpoint. Let $\mathcal{A}_{ns}$ contain the foreign origins with tariffs observed at both endpoints, whether or not their output-price change is observed. Using this set at each endpoint, the imputation is

$$\bar{\mathcal{T}}_{ns,t} = \frac{\sum_{i \in \mathcal{A}_{ns}} B_{nis,t} \mathcal{T}_{nis,t}}{\sum_{i \in \mathcal{A}_{ns}} B_{nis,t}},$$

$$X_{n,ROW,s,t} = (1 + \bar{\mathcal{T}}_{ns,t}) B_{n,ROW,s,t}.$$

We assign this tariff separately in each year. This allows us to include ROW in total foreign expenditure, but we still lack observed tariff and output-price changes for ROW. We therefore construct the shifters using only origins with observed tariffs and prices and rescale their weights to sum to one. In the all-goods samples, the median share of initial foreign expenditure represented by these origins is 91.72 percent in WIOD and 98.37 percent in OECD; 36 and 28 observations, respectively, have coverage below 50 percent. In the manufacturing subsets, the corresponding medians are 93.14 and 98.94 percent.

Tariff Concordances. Both data sets use the ISIC industry-level `tariff95` measure from the Global Tariff Database (Teti, 2024). When several tariff industries correspond to one input-output sector, we combine their tariffs separately for each origin-destination pair. In both data sets, we require tariffs for every component industry to be observed in both years; otherwise, we treat the sector’s tariff change as missing. We average tariff levels with equal weights across all component industries in each year before calculating the change in the log gross tariff. In OECD, chemicals and pharmaceuticals share ISIC sector 24, ferrous and non-ferrous basic metals share sector 27, and ships and other non-motor-vehicle transport equipment share sector 35. The OECD concordance covers all nineteen manufacturing sectors and seven non-manufacturing goods sectors. WIOD uses the corresponding concordance to its fourteen manufacturing sectors and two non-manufacturing goods sectors.

Trade-Agreement Tariff Instruments. We identify agreement membership and implementation dates using the WTO Regional Trade Agreements Database and official institutional sources (World Trade Organization, 2025). The tariff instrument uses the 1995 and 2007 `tariff95` data, with the same industry concordances as the endogenous tariff shifter. These targets summarize broad liberalization commitments and abstract from product-specific exceptions and phase-in schedules. In particular, the EU-Türkiye targets apply across the goods sectors, including products outside the customs union’s coverage. Similarly, the NAFTA targets impose zero tariffs even on products for which tariff elimination was scheduled for 2008.

For the ten countries that joined the EU in May 2004, we set the target tariff to zero on imports from the other countries in the enlarged EU25. For imports from outside the EU25, we use the average 2007 tariff across EU15 countries for the same origin and sector. For EU15 destinations, we set the target tariff to zero on imports from the ten entrants. We construct the bilateral instrument as the change in the log gross tariff from the observed 1995 tariff to this target.

We also include the 2007 EU enlargement, the EU-Türkiye customs union, the Gulf Cooperation Council, and the West African Economic and Monetary Union. For Bulgaria, Romania, and Türkiye, we set target tariffs to zero on imports from EU27 members. For imports from outside the EU27, we use the average 2007 tariff across EU15 countries for the same origin and sector. For EU15 destinations, we also set target tariffs to zero on imports from Bulgaria and Romania. The EU-Türkiye component includes only Türkiye as a destination because the EU had already eliminated tariffs on Turkish industrial goods before 1995. For the Gulf Cooperation Council and the West African Economic and Monetary Union, we set target tariffs to zero on imports from other union members. For imports in a sector s into a destination d that is in a given union coming from an origin country o that is outside that union, we use the average of the 2007 tariffs applied to the same os pair across the other observed union members $u \neq d$, thereby excluding the tariff applied by the destination itself.

We also include MERCOSUR, setting target tariffs to zero on imports from other included members and using the average 2007 tariff across the other included members for each external origin and sector. The GCC calculation uses Saudi Arabia and the United Arab Emirates, the WAEMU calculation uses Côte d'Ivoire and Senegal, and the MERCOSUR calculation uses Argentina and Brazil.

For NAFTA, PAFTA, and the Common Economic Zone, we set target tariffs to zero on trade among included members. We also set target tariffs to zero in both directions for Chile–Mexico, Israel–Mexico, New Zealand–Singapore, Singapore–Australia, Türkiye–Israel, EFTA–Singapore, EU–Israel, and EU–Croatia. The EU agreements use EU27 membership, and EFTA includes Iceland, Norway, and Switzerland. For AFTA, we include trade among Brunei, Indonesia, Malaysia, the Philippines, Singapore, Thailand, and Vietnam, and set the target tariff to the smaller of the observed 1995 tariff and 5 percent. We exclude Cambodia, Laos, and Myanmar from this component of the instrument because their schedules for reducing tariffs to 0–5 percent extended beyond 2007. If a pair is covered by both a customs union and one of these preferential agreements, we use the target tariff from the preferential agreement.

We set the instrument to zero for bilateral pairs outside these agreements. The applied tariff shifter includes observed tariff changes for all origins used to construct the shifters, including those outside these agreements.

Consumer and Producer Manufacturing Allocation. We construct the allocations separately in WIOD and OECD using initial-year basic-price absorption, including ROW. For manufacturing sector s , let

$$\chi_s = \frac{\text{world final consumption of sector } s}{\text{world absorption of sector } s}.$$

Final consumption includes each of the following: household, nonprofit, and government purchases of domestic and foreign goods. Total absorption additionally includes the following: intermediate uses, investment, and inventories. We classify manufacturing sectors with values of χ_s at or above the manufacturing median as consumer-facing. We classify the remaining sectors as producer-facing. The median value of χ_s is 20.22 percent in WIOD and 16.58 percent in OECD. We keep these classifications fixed across changes in the sample, tariff measure, and control specification.

Table 5: WIOD Goods-Sector Allocation

Code	Sector	Group
15t16	Food, Beverages and Tobacco	Consumer
17t18	Textiles and Textile Products	Consumer
19	Leather, Leather and Footwear	Consumer
20	Wood and Products of Wood and Cork	Producer
21t22	Pulp, Paper, Paper , Printing and Publishing	Producer
23	Coke, Refined Petroleum and Nuclear Fuel	Consumer
24	Chemicals and Chemical Products	Consumer
25	Rubber and Plastics	Producer
26	Other Non-Metallic Mineral	Producer
27t28	Basic Metals and Fabricated Metal	Producer
29	Machinery, Nec	Producer
30t33	Electrical and Optical Equipment	Producer
34t35	Transport Equipment	Consumer
36t37	Manufacturing, Nec; Recycling	Consumer
AtB	Agriculture, hunting, forestry and fishing	Non-manufacturing
C	Mining and quarrying	Non-manufacturing

Table 6: OECD ICIO Goods-Sector Allocation

Code	Sector	Group
C10T12	Food, beverages, tobacco	Consumer
C13T15	Textiles, apparel, leather	Consumer
C16	Wood products	Producer
C17_18	Paper and printing	Producer
C19	Coke and refined petroleum	Consumer
C20	Chemicals	Consumer
C21	Pharmaceuticals	Consumer
C22	Rubber and plastics	Producer
C23	Other non-metallic minerals	Producer
C24A	Iron and steel	Producer
C24B	Non-ferrous metals	Producer
C25	Fabricated metals	Producer
C26	Computer, electronic, optical	Consumer
C27	Electrical equipment	Consumer
C28	Machinery and equipment	Producer
C29	Motor vehicles	Consumer
C301	Ships and boats	Producer
C302T309	Rail, aircraft, other transport	Consumer
C31T33	Furniture, other manufacturing, repair	Consumer
A01	Agriculture and hunting	Non-manufacturing
A02	Forestry and logging	Non-manufacturing
A03	Fishing and aquaculture	Non-manufacturing
B05	Mining of coal and lignite	Non-manufacturing
B06	Extraction of oil and gas	Non-manufacturing
B07	Mining of metal ores	Non-manufacturing
B08	Other mining and quarrying	Non-manufacturing

A.2 Numerical Implementation of the Baseline Estimator

Linear Tails of the Restricted Cubic Spline. To see that the spline is linear below the first knot, note that $R(\lambda) = 0$ for all $\lambda < a$, in which case the elasticity is simply $\varepsilon_s(\lambda) = \theta_{s0} + \theta_{s1}\lambda$. Similarly, above the final knot, $R(\lambda) = \frac{b-a}{c-a} [3\lambda - (a + b + c)]$ for all $\lambda > c$, in which case the elasticity is

$$\varepsilon_s(\lambda) = \left[\theta_{s0} - \theta_{s2} \frac{b-a}{c-a} (a + b + c) \right] + \left[\theta_{s1} + 3\theta_{s2} \frac{b-a}{c-a} \right] \lambda,$$

which is also linear in the domestic expenditure share.

Two-Stage Least Squares. We first calculate the three knots within each group and construct $R_g(\lambda_{nns,0})$ from the formula in Section 4.1. We then create sector interactions with Q_{ns}^T and group interactions with $\lambda_{nns,0}Q_{ns}^T$ and $R_g(\lambda_{nns,0})Q_{ns}^T$. Their instruments replace Q_{ns}^T with Z_{ns}^T . The included additive controls are group intercepts, domestic-share terms, and spline terms. The measured-price controls contain sector interactions with Q_{ns}^P and group

interactions with $\lambda_{nns,0}Q_{ns}^P$ and $R_g(\lambda_{nns,0})Q_{ns}^P$. Estimation is ordinary linear 2SLS with these constructed variables.

To give the matrix expression, let D be the matrix of tariff regressors, W the controls, Z the excluded instruments, and $\mathbf{y}$ the outcome vector. Let M_W remove the linear projection on W , and define $\tilde{D} = M_W D$, $\tilde{Z} = M_W Z$, and $\tilde{\mathbf{y}} = M_W \mathbf{y}$. Then

$$\begin{aligned}\hat{\boldsymbol{\theta}} &= (\tilde{D}' P_{\tilde{Z}} \tilde{D})^{-1} \tilde{D}' P_{\tilde{Z}} \tilde{\mathbf{y}}, \\ P_{\tilde{Z}} &= \tilde{Z}(\tilde{Z}' \tilde{Z})^{-1} \tilde{Z}'.\end{aligned}$$

The control coefficients are obtained by regressing $\mathbf{y} - D\hat{\boldsymbol{\theta}}$ on W . We calculate the covariance matrix using the full regression. Conditional on the domestic shares and knot locations, each reported slope and fitted elasticity is a linear combination of the estimated coefficients. We calculate its standard error from the full destination-clustered covariance matrix using the delta method. With baseline controls, the groups have separate coefficients and can be estimated separately. Their standard errors still allow dependence across a destination's sectors. When country-characteristic coefficients are common across groups, we estimate the groups jointly.

We multiply the destination-clustered sandwich covariance matrix by $G/(G-1)$, where G is the number of destination countries, and use $G-1$ degrees of freedom for reported significance comparisons. The conditional SW calculations use each group's observations and instrument count, with multiplier $[G/(G-1)][(N_g-1)/(N_g-L_g)]$, where L_g counts all instruments including controls. Exactly identified conditional tests use reference degrees of freedom $(1, G-1)$. With common country-characteristic coefficients, the reported minimum is over the joint regression. Coefficients, covariance matrices, and conditional first-stage statistics were checked against an independent implementation in `ivreg2`. These calculations treat published prices, shares, weights, and knot locations as observed.

Alternative Specifications. The country-characteristics exercise adds the normalized characteristics to the elasticity intercept and measured-price coefficient, with separate common coefficients. The sector-specific-control exercise allows the additive-control intercept to vary by sector, while keeping the share-dependent coefficients and knot locations group-specific. Existing group knots remain fixed in the coverage, tariff, and omitted-observation exercises.

The trade-weighted tariff exercise constructs Q_{ns}^T using the available trade-weighted applied-tariff series and renormalizes its origin weights over matched tariffs. The outcome, initial share, price control, and agreement instrument retain their baseline definitions. The quadratic exercise replaces $R_g(\lambda)$ by λ^2 in the elasticity and both control functions.

Lagged-Share Instruments. The measurement-error exercise uses 1996–2007 expenditure, tariff, and output-price changes, retaining 1995 foreign-origin weights and the 1995–2007 agreement shifter. Knot locations retain their baseline values. In the same-period comparison, 1996 domestic shares enter both regressors and instrument interactions. In the

lagged version, for each of the two functions $f_g(\lambda) = \lambda$ and $f_g(\lambda) = R_g(\lambda)$, we instrument

$$\begin{aligned} f_g(\lambda_{nns,1996}) & \text{ with } f_g(\lambda_{nns,1995}), \\ f_g(\lambda_{nns,1996})Q_{ns}^{\mathcal{T}} & \text{ with } f_g(\lambda_{nns,1995})Z_{ns}^{\mathcal{T}}, \\ f_g(\lambda_{nns,1996})Q_{ns}^P & \text{ with } f_g(\lambda_{nns,1995})Q_{ns}^P. \end{aligned}$$

All interactions remain specific to their group. Sector tariff terms use sector interactions with $Z_{ns}^{\mathcal{T}}$; group constants and sector-level Q_{ns}^P remain included controls.

Constrained and Finite-Change Estimates. The constrained estimator minimizes the same 2SLS criterion while requiring

$$\theta_{s0} + \theta_{g(s)1}\lambda + \theta_{g(s)2}R_{g(s)}(\lambda) \geq 0.1 \quad \text{for every } s \text{ and every } \lambda \in [0, 1].$$

We first remove the included controls by linear projection and then estimate the elasticity coefficients subject to this restriction. We initially impose the bound at a finite set of domestic expenditure shares. If the fitted function falls below the bound between these shares, we impose the bound at the share where this minimum occurs and estimate again. We repeat this procedure until the entire function satisfies the bound. To check this, we evaluate the function at zero and one, at each knot, and wherever its derivative is zero within each cubic segment. Checking only the endpoints would not be sufficient because the function can reach its minimum between them. We also check the first-order conditions for the constrained estimates. Imposing the bound leaves the first-stage regressions unchanged.

For the finite-change exercise, let $\ell_{ns}(0) = \ln[\lambda_{nns,0}/(1 - \lambda_{nns,0})]$ and solve

$$\frac{d\ell_{ns}(v)}{dv} = Q_{ns}^{\mathcal{T}}\varepsilon_s \left(\frac{\exp \ell_{ns}(v)}{1 + \exp \ell_{ns}(v)} \right), \quad 0 \leq v \leq 1.$$

This allows the elasticity to change as the domestic expenditure share changes in response to tariffs. The resulting tariff response is $F_{ns}(\boldsymbol{\theta}) = \ell_{ns}(1) - \ell_{ns}(0)$, which replaces $\varepsilon_s(\lambda_{nns,0})Q_{ns}^{\mathcal{T}}$ in equation (19). We continue to evaluate the additive and measured-price controls at the initial share. Using the same excluded instruments, we minimize

$$[M_W\{\mathbf{y} - F(\boldsymbol{\theta})\}]' P_{\tilde{Z}} [M_W\{\mathbf{y} - F(\boldsymbol{\theta})\}]$$

subject to the same lower bound of 0.1 on the entire elasticity function. For each candidate set of elasticity coefficients, we estimate the control coefficients by linear projection. We start the numerical procedure from four sets of coefficients that satisfy the bound and retain the solution that satisfies the bound and gives the lowest value of the criterion. We check the constrained first-order conditions and verify that the entire function satisfies the bound. We also check the numerical integration using an independent differential-equation solver. These checks do not guarantee that we have found the global minimum.

For the unrestricted finite-change estimates, we minimize the same criterion without the lower bound. We calculate destination-clustered standard errors using the derivatives of the nonlinear regression.

This exercise allows the elasticity to vary over the tariff change while keeping the control functions fixed at their initial-share values. It therefore relaxes the local approximation to

the tariff response. It retains the approximation used to combine tariff changes across origins and the assumed form of the controls. Allowing tariffs, output prices, and demand shifts to change jointly along a specified path would require additional assumptions. We report only point estimates for both constrained exercises.

A.3 Additional Empirical Results

The following tables use WIOD and the three-knot spline unless otherwise stated. As in the baseline, we report the derivative evaluated at each sector's median domestic expenditure share and averaged across sectors, the share of observations with nonnegative fitted elasticities, and elasticities evaluated at each sector's quartiles. We report these same statistics when comparing alternative functional forms. For the constrained estimates, we report only point estimates. The non-manufacturing estimates are identical across Specifications (2) and (3) when all coefficients are estimated separately by group. They can differ when country-characteristic coefficients are common across groups.

Table 7: WIOD Estimates with Country Characteristics in Elasticity and Measured-Price Coefficients

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	13.07 (5.77)	2.79 (1.92)	6.60 (2.21)	77.5%	561.25	627
(2)	Manufacturing	14.04 (6.62)	3.37 (2.09)	7.53 (2.73)	80.4%	519.86	547
	Non-manuf. goods	29.27 (15.00)	-1.42 (5.30)	6.23 (4.73)	63.7%	519.86	80
(3)	Consumer manuf.	15.37 (7.23)	1.08 (2.66)	5.25 (3.42)	66.9%	717.91	269
	Producer manuf.	35.34 (11.31)	3.27 (2.99)	13.38 (3.37)	83.1%	717.91	278
	Non-manuf. goods	34.84 (15.37)	-2.53 (5.36)	6.42 (4.79)	58.8%	717.91	80

Notes: This table modifies Table 1 by allowing destination characteristics to enter the elasticity intercept and the coefficient on measured-price changes, with coefficients common across groups. The reported quartile elasticities also average the characteristic contribution across destinations within each sector. Min. SW F is calculated for the entire joint regression and therefore repeats across its group rows. Non-manufacturing estimates can differ across Specifications (2) and (3).

Table 8: WIOD Estimates with Sector-Specific Control Intercepts

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	14.16 (6.57)	3.25 (1.14)	7.36 (2.13)	79.7%	874.49	627
(2)	Manufacturing	15.15 (7.74)	3.70 (1.37)	8.12 (2.59)	81.4%	695.00	547
	Non-manuf. goods	31.77 (16.92)	-1.01 (4.77)	8.87 (8.16)	72.5%	15.11	80
(3)	Consumer manuf.	18.90 (8.36)	0.84 (1.92)	5.85 (3.02)	66.2%	398.72	269
	Producer manuf.	27.79 (13.88)	4.35 (1.78)	12.61 (3.94)	85.3%	180.26	278
	Non-manuf. goods	31.77 (16.92)	-1.01 (4.77)	8.87 (8.16)	72.5%	15.11	80

Notes: This table modifies Table 1 by allowing the additive-control intercept to vary by sector rather than group. The elasticity and measured-price-control intercepts remain sector-specific, and all share-dependent coefficients remain group-specific.

Table 9: WIOD Estimates with At Least 50 Percent Initial Import Coverage

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	12.68 (4.85)	3.28 (1.13)	7.31 (1.76)	84.3%	3,910.52	591
(2)	Manufacturing	13.86 (5.93)	3.52 (1.19)	7.86 (1.97)	84.9%	2,163.54	535
	Non-manuf. goods	21.25 (15.51)	-0.33 (6.11)	4.78 (3.00)	76.8%	6,465.13	56
(3)	Consumer manuf.	16.45 (6.99)	1.02 (1.53)	5.70 (2.22)	70.8%	1,173.88	260
	Producer manuf.	24.17 (8.20)	4.59 (2.00)	11.81 (2.76)	87.6%	300.94	275
	Non-manuf. goods	21.25 (15.51)	-0.33 (6.11)	4.78 (3.00)	76.8%	6,465.13	56

Notes: This table replicates Table 1 after requiring the named foreign origins entering the shifters to account for at least half of total initial foreign expenditure. Baseline group knot locations and sector classifications are held fixed.

Table 10: WIOD Estimates with Quadratic Elasticity Schedules and Controls

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	13.83 (4.84)	3.15 (1.25)	7.02 (1.81)	82.6%	4,978.40	627
(2)	Manufacturing	16.01 (6.44)	3.49 (1.33)	7.96 (2.23)	85.4%	2,031.77	547
	Non-manuf. goods	22.73 (11.51)	-1.32 (4.63)	5.32 (4.13)	72.5%	221.68	80
(3)	Consumer manuf.	17.31 (7.29)	0.81 (1.92)	5.28 (2.67)	68.4%	1,019.75	269
	Producer manuf.	24.25 (7.45)	4.49 (1.95)	11.68 (2.50)	85.6%	293.80	278
	Non-manuf. goods	22.73 (11.51)	-1.32 (4.63)	5.32 (4.13)	72.5%	221.68	80

Notes: This table modifies Table 1 by replacing the spline with a quadratic in the elasticity schedule, additive controls, and coefficient on measured-price changes. The coefficient structure otherwise remains unchanged.

Table 11: WIOD Estimates with a Positive Elasticity Floor

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	N
(1)	All goods	4.81	4.89	6.05	100.0%	627
(2)	Manufacturing	5.66	5.35	6.76	100.0%	547
	Non-manuf. goods	6.17	1.05	2.09	100.0%	80
(3)	Consumer manuf.	3.82	3.36	4.35	100.0%	269
	Producer manuf.	12.89	7.38	10.32	100.0%	278
	Non-manuf. goods	6.17	1.05	2.09	100.0%	80

Notes: This table replicates Panel A of Table 1, imposing $\varepsilon_s(\lambda) \geq 0.1$ for every sector and every $\lambda \in [0, 1]$. Only point estimates are reported; no constrained standard errors or confidence intervals are reported.

Table 12: WIOD Estimates with Finite Tariff Responses

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	N
(1)	All goods	13.30	4.12	8.35	83.6%	627
		(6.04)	(1.56)	(2.35)		
(2)	Manufacturing	16.60	4.62	9.72	85.4%	547
		(9.58)	(1.75)	(3.25)		
	Non-manuf. goods	24.28	-1.79	5.33	71.2%	80
		(16.35)	(5.66)	(4.19)		
(3)	Consumer manuf.	23.53	0.86	7.30	68.0%	269
		(14.42)	(2.44)	(4.11)		
	Producer manuf.	33.97	25.58	40.78	79.5%	278
		(83.19)	(94.77)	(135.93)		
	Non-manuf. goods	24.28	-1.79	5.33	71.2%	80
		(16.35)	(5.66)	(4.19)		

Notes: This table modifies Panel A of Table 1 by integrating the tariff response along the elasticity schedule, without imposing a positive lower bound. Additive and measured-price controls remain evaluated at the initial share. Parentheses contain destination-clustered standard errors from the nonlinear IV estimator. Conditional SW statistics are not reported for this estimator.

Table 13: WIOD Estimates with a Positive Elasticity Floor and Finite Tariff Responses

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	N
(1)	All goods	4.35	5.00	6.06	100.0%	627
(2)	Manufacturing	5.24	5.50	6.81	100.0%	547
	Non-manuf. goods	6.46	1.03	2.13	100.0%	80
(3)	Consumer manuf.	3.28	3.42	4.29	100.0%	269
	Producer manuf.	7.12	7.98	9.67	100.0%	278
	Non-manuf. goods	6.46	1.03	2.13	100.0%	80

Notes: This table modifies Panel A of Table 1 by integrating the tariff response along the elasticity schedule and imposing $\varepsilon_s(\lambda) \geq 0.1$ for every sector and every $\lambda \in [0, 1]$. Additive and measured-price controls remain evaluated at the initial share. Only point estimates are reported; conditional SW statistics and constrained standard errors and confidence intervals are not reported.

Table 14: OECD ICIO Estimates with a Positive Elasticity Floor and Finite Tariff Responses

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	N
(1)	All goods	2.88	5.00	5.77	100.0%	2,030
(2)	Manufacturing	0.00	3.79	3.79	100.0%	1,510
	Non-manuf. goods	15.26	8.23	11.62	100.0%	520
(3)	Consumer manuf.	2.23	3.65	4.39	100.0%	793
	Producer manuf.	0.00	3.92	3.92	100.0%	717
	Non-manuf. goods	15.26	8.23	11.62	100.0%	520

Notes: This table modifies Panel A of Table 2 by integrating the tariff response along the elasticity schedule and imposing $\varepsilon_s(\lambda) \geq 0.1$ for every sector and every $\lambda \in [0, 1]$. Additive and measured-price controls remain evaluated at the initial share. Only point estimates are reported; conditional SW statistics and constrained standard errors and confidence intervals are not reported.

Table 15: WIOD Estimates Instrumenting Measured-Price Changes

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
<i>A. Three-knot spline</i>							
(1)	All goods	10.57 (6.24)	3.11 (1.59)	6.69 (2.31)	72.2%	4,703.11	627
(2)	Manufacturing	13.13 (7.58)	3.43 (1.67)	7.73 (2.59)	81.2%	2,791.38	547
	Non-manuf. goods	19.13 (28.10)	3.95 (10.40)	9.07 (6.50)	95.0%	3.68	80
(3)	Consumer manuf.	19.47 (12.76)	1.38 (2.57)	6.84 (4.28)	74.3%	64.38	269
	Producer manuf.	21.94 (10.49)	5.00 (2.14)	11.78 (3.59)	87.4%	24.13	278
	Non-manuf. goods	19.13 (28.10)	3.95 (10.40)	9.07 (6.50)	95.0%	3.68	80
<i>B. Linear schedule</i>							
(1)	All goods	13.86 (7.82)	3.39 (1.49)	7.43 (2.59)	80.4%	2,331.09	627
(2)	Manufacturing	15.70 (8.87)	3.87 (1.66)	8.55 (3.12)	83.7%	1,087.86	547
	Non-manuf. goods	-0.75 (21.42)	6.60 (9.50)	6.41 (5.82)	100.0%	9.32	80
(3)	Consumer manuf.	10.34 (11.62)	2.03 (2.67)	5.52 (4.12)	75.5%	13.83	269
	Producer manuf.	35.70 (14.17)	4.72 (2.08)	13.95 (4.45)	87.8%	63.61	278
	Non-manuf. goods	-0.75 (21.42)	6.60 (9.50)	6.41 (5.82)	100.0%	9.32	80

Notes: This table replicates Table 1, instrumenting Q_{ns}^P and all its interactions as well as the tariff terms. For each country and sector, we average observed output-price changes in USD equally across the other fifteen goods sectors. We construct the price instrument using the baseline foreign-origin weights and subtract the home-country average. Both instruments have the same sector and share interactions as their corresponding regressors. The samples, additive controls, tariff instruments, and spline knots are unchanged. The averages always use all fifteen other goods sectors, including in Specifications (2) and (3). Slopes and quartile elasticities are defined as in Table 1. Share ≥ 0 is the percentage of observations with a non-negative fitted elasticity. Destination-clustered standard errors are in parentheses. Min. SW F reports the smallest conditional first-stage statistic across both tariff and price terms within the row's group, using destination-clustered variances.

Table 16: WIOD Estimates without Measured-Price Controls

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
<i>A. Three-knot spline</i>							
(1)	All goods	10.88 (4.64)	3.22 (1.21)	6.93 (2.18)	79.9%	28,137.59	627
(2)	Manufacturing	10.85 (5.63)	3.59 (1.23)	7.33 (2.34)	82.4%	5,450.43	547
	Non-manuf. goods	30.46 (12.96)	-4.03 (3.37)	4.77 (5.19)	66.2%	197.93	80
(3)	Consumer manuf.	15.79 (7.44)	1.94 (1.72)	6.73 (2.72)	76.6%	2,487.65	269
	Producer manuf.	19.38 (10.23)	4.23 (1.90)	10.05 (2.99)	87.8%	538.51	278
	Non-manuf. goods	30.46 (12.96)	-4.03 (3.37)	4.77 (5.19)	66.2%	197.93	80
<i>B. Linear schedule</i>							
(1)	All goods	13.80 (6.29)	3.26 (1.28)	7.29 (2.44)	86.1%	57,096.98	627
(2)	Manufacturing	13.82 (6.86)	3.71 (1.38)	7.83 (2.73)	86.5%	10,437.15	547
	Non-manuf. goods	25.55 (10.20)	-3.34 (3.33)	3.06 (3.03)	58.8%	525.70	80
(3)	Consumer manuf.	12.27 (7.57)	2.80 (1.91)	6.93 (2.99)	83.6%	4,104.58	269
	Producer manuf.	28.38 (12.14)	3.72 (1.91)	11.06 (3.57)	87.4%	765.05	278
	Non-manuf. goods	25.55 (10.20)	-3.34 (3.33)	3.06 (3.03)	58.8%	525.70	80

Notes: This table replicates Table 1, omitting Q_{ns}^P and all its interactions. The samples, additive controls, tariff instruments, and spline knots are unchanged. Panel A uses the three-knot spline and Panel B the linear specification. Slopes are evaluated at each sector's median initial domestic share and averaged equally across sectors; linear slopes are constant. $\hat{\varepsilon}_p$ evaluates each sector at its own p th share percentile and averages equally across sectors. Share ≥ 0 reports the percentage of observations with non-negative fitted elasticities. Destination-clustered standard errors are in parentheses; Min. SW F is the smallest conditional first-stage statistic within the row's group.

Table 17: WIOD Estimates with 1996–2007 Changes with Current-Share Interactions

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	11.09 (3.26)	2.21 (1.17)	6.42 (1.44)	73.2%	2,193.45	627
(2)	Manufacturing	10.86 (4.19)	2.52 (1.38)	6.81 (1.61)	72.6%	1,993.03	547
	Non-manuf. goods	25.26 (18.63)	1.61 (5.47)	9.01 (9.57)	82.5%	53.97	80
(3)	Consumer manuf.	14.08 (5.72)	-0.18 (1.99)	4.92 (1.65)	63.2%	684.42	269
	Producer manuf.	25.24 (8.31)	4.33 (1.68)	11.97 (2.77)	84.9%	114.76	278
	Non-manuf. goods	25.26 (18.63)	1.61 (5.47)	9.01 (9.57)	82.5%	53.97	80

Notes: This table modifies Table 1 by measuring outcomes and observed tariff and price changes over 1996–2007. Regressors and instrument interactions use 1996 domestic shares. Origin weights and knots retain their 1995 definitions, and the agreement instrument retains its 1995–2007 definition.

Table 18: WIOD Estimates with 1996–2007 Changes with Lagged-Share Instruments

Spec.	Aggregate	Slope at medians	$\hat{\varepsilon}_{25}$	$\hat{\varepsilon}_{75}$	Share ≥ 0	Min. SW F	N
(1)	All goods	11.13 (3.53)	2.42 (1.23)	6.15 (1.46)	73.5%	1,471.83	627
(2)	Manufacturing	11.52 (4.52)	2.69 (1.47)	6.63 (1.67)	74.8%	736.14	547
	Non-manuf. goods	20.83 (19.24)	2.12 (5.73)	7.46 (9.31)	86.2%	35.40	80
(3)	Consumer manuf.	14.47 (5.83)	-0.18 (2.19)	4.32 (1.89)	57.6%	324.28	269
	Producer manuf.	21.80 (7.84)	4.63 (1.71)	11.28 (2.58)	85.3%	171.63	278
	Non-manuf. goods	20.83 (19.24)	2.12 (5.73)	7.46 (9.31)	86.2%	35.40	80

Notes: This table modifies Table 1 by constructing every share-dependent tariff, additive, and measured-price term with the 1996 domestic share and instrumenting it with its 1995-share counterpart. Origin weights and knots are dated 1995, and the agreement instrument retains its 1995–2007 definition. Min. SW F covers all instrumented regressors, including the additive and measured-price share terms.

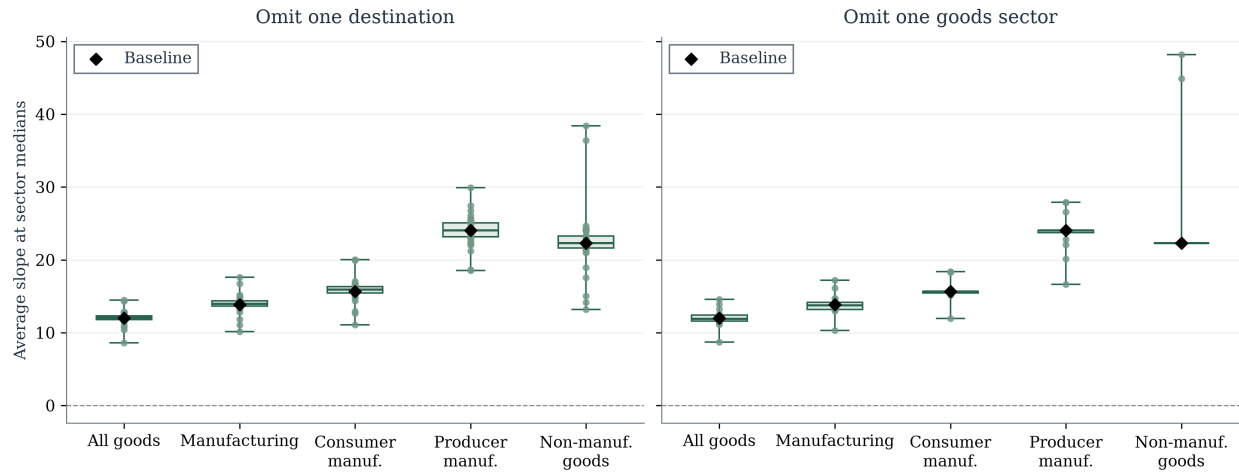


Figure 4: WIOD Leave-One-Out Estimates of Home-Foreign Elasticity Slopes at Sector Medians

Notes: The left panel omits each destination country in turn; the right panel omits each goods sector in turn. Knots and sector classifications retain their baseline values. The figure shows slopes, defined exactly as in Table 1. Boxes show the interquartile range, lines show medians, whiskers show the full range, and diamonds mark baseline estimates. The non-manufacturing group contains only two sectors.

Theoretical Appendix

B Proofs

B.1 Proof of Proposition 1

Because utility is homogeneous of degree one, a consumer attaining utility W_n has Hicksian demand $W_n h_{ni}(\mathbf{p}_n)$ and total expenditure $W_n e_n(\mathbf{p}_n)$. Homogeneity of Hicksian demand and the expenditure function therefore gives

$$\lambda_{ni}(t) = \frac{\pi_{ni}(t) h_{ni}(\boldsymbol{\pi}_n(t))}{H_n(t)}.$$

Differentiating $H_n(t) = e_n(\boldsymbol{\pi}_n(t))$ and applying Shephard's lemma yields

$$\frac{H'_n(t)}{H_n(t)} = \sum_{i \neq n} \lambda_{ni}(t) \frac{d \ln \pi_{ni}(t)}{dt} = [1 - \lambda_{nn}(t)] z'_n(t),$$

where we used $\pi_{nn}(t) = 1$ and the definition of $z_n(t)$. This gives equation (2). Since $Y_n/p_{nn} = a_n L_n$ at each equilibrium endpoint, $W_n^1/W_n^0 = H_n(t_0)/H_n(t_1)$. Integrating the previous expression between t_0 and t_1 gives equation (4).

If, in addition, $z'_n(t) > 0$ and $\varepsilon_n^\Gamma(t) > 0$ throughout the relevant interval, the domestic expenditure share is strictly increasing and can index the path. Since $X_{nn}/X_{nF} = \lambda_{nn}/(1 - \lambda_{nn})$, the definition of the home-foreign elasticity gives

$$d \ln \lambda_{nn}(t) = [1 - \lambda_{nn}(t)] \varepsilon_n^\Gamma(t) dz_n(t).$$

Combining this expression with equation (2) yields

$$d \ln H_n(t) = \frac{d \ln \lambda_{nn}(t)}{\varepsilon_n^\Gamma(\lambda_{nn}(t))}.$$

Changing variables from t to λ_{nn} and integrating between the two equilibria gives equation (5).

B.2 Proof of Corollary 1

Let H_n^A denote the finite quantity of the home good required to attain one unit of utility using that good alone. This bundle is feasible in the expenditure minimization problem at every t , so $H_n(t) \leq H_n^A$. Expenditure on each good is bounded by total expenditure, so $0 \leq \pi_{ni}(t) h_{ni}(\boldsymbol{\pi}_n(t)) \leq H_n(t) \leq H_n^A$. Dividing by $\pi_{ni}(t)$ and taking the limit gives

$$0 \leq h_{ni}(\boldsymbol{\pi}_n(t)) \leq \frac{H_n^A}{\pi_{ni}(t)} \rightarrow 0 \quad \text{for every } i \neq n.$$

Continuity of preferences and the unit-utility constraint then imply $h_{nn}(\boldsymbol{\pi}_n(t)) \rightarrow H_n^A$. Since $h_{nn}(\boldsymbol{\pi}_n(t)) \leq H_n(t) \leq H_n^A$, we obtain $H_n(t) \rightarrow H_n^A$ and $\lambda_{nn}(t) = h_{nn}(\boldsymbol{\pi}_n(t))/H_n(t) \rightarrow 1$.

Income in units of the home good equals $a_n L_n$ at both the initial trade equilibrium and autarky. Consequently, $W_n^A = a_n L_n / H_n^A > 0$ and

$$\ln \frac{W_n^T}{W_n^A} = \ln \frac{H_n^A}{H_n(1)}.$$

Integrating equation (2) from 1 to a finite t and taking $t \rightarrow \infty$ gives equation (6). Under the additional positivity conditions, the change of variables established in the proof of Proposition 1 gives equation (7). Every path satisfying the stated conditions gives the same gains from trade because the integral equals the change in log unit expenditure between the same endpoints.

B.3 Identification with Instrumented Price Changes

Here, we provide conditions for identifying the elasticity when both the tariff and measured-price components are instrumented. Let $y_{ns} \equiv \Delta \ln(X_{nns}/X_{nFs})$ and let V_{ns} collect predetermined shares and other controls, excluding Q_{ns}^P . Consider the estimating relationship

$$y_{ns} = \varepsilon_s(\lambda_{nns,0})Q_{ns}^T + \gamma_s(\lambda_{nns,0})Q_{ns}^P + a_s(\lambda_{nns,0}, V_{ns}) + v_{ns}.$$

As in equation (19), the coefficient on the tariff component is the home-foreign elasticity, while the measured-price component has a separate coefficient function. This specification allows measured prices to differ from the production prices in the model, provided that the resulting residual satisfies the exclusion restriction below. The function a_s captures additive controls. We impose no functional form on ε_s , γ_s , or a_s .

Let $\mathbf{Q}_{ns} = (Q_{ns}^T, Q_{ns}^P)'$ collect the two price components and let $\mathbf{Z}_{ns} = (Z_{ns}^T, Z_{ns}^P)'$ collect their instruments.

Proposition 3. *Fix a sector s and an interval $\mathcal{L}_s \subset (0, 1)$ contained in the support of its initial domestic expenditure shares. Suppose the estimating relationship above holds, ε_s and γ_s are continuous on $\mathcal{L}_s$, and all conditional second moments below are finite. Conditional on $(\lambda_{nns,0} = \lambda, V_{ns}, s)$, suppose the exclusion restriction*

$$\text{Cov}(\mathbf{Z}_{ns}, v_{ns} \mid \cdot) = \mathbf{0}$$

and the relevance condition

$$\text{rank Cov}(\mathbf{Z}_{ns}, \mathbf{Q}_{ns} \mid \cdot) = 2$$

hold for almost every value of $(\lambda_{nns,0}, V_{ns})$ with $\lambda_{nns,0} \in \mathcal{L}_s$. Then ε_s and γ_s are identified on $\mathcal{L}_s$. At almost every such conditioning value, they satisfy

$$\begin{pmatrix} \varepsilon_s(\lambda) \\ \gamma_s(\lambda) \end{pmatrix} = \text{Cov}(\mathbf{Z}_{ns}, \mathbf{Q}_{ns} \mid \cdot)^{-1} \text{Cov}(\mathbf{Z}_{ns}, y_{ns} \mid \cdot).$$

Proof. Take the conditional covariance of the estimating relationship with each instrument. The additive term is fixed by the conditioning variables, and the exclusion restriction removes the residual. This gives two equations in $\varepsilon_s(\lambda)$ and $\gamma_s(\lambda)$. The relevance condition makes their coefficient matrix invertible, giving the displayed expression. Continuity extends identification from almost every share to all of $\mathcal{L}_s$. $\square$

C An Armington Gravity Model

Here, we restrict preferences to a parametric expenditure system. Our goal is to provide a simple general-equilibrium gravity model that strictly generalizes the canonical CES framework. The substantive generalization is in the upper nest between the home good and the foreign aggregate. We retain CES aggregation across foreign sources but replace the CES upper nest with a constant-difference-of-elasticities (CDE) expenditure system (Hanoch, 1975). The CES restriction generates a constant home-foreign elasticity. The CDE upper nest generates the linear special case of equation (16), with the coefficient on the spline term set to zero.

A non-nested CDE system over national varieties could also be used to construct a gravity model with variable substitution elasticities. We use the nested specification because one of our focuses is the distinction between substitution across foreign sources and substitution between the home good and the foreign aggregate. The nested system allows the home-foreign elasticity function to vary separately from the elasticity across foreign sources.

Sections C.1 and C.2 develop the one-sector version, corresponding to a simplification of Section 2. Section C.3 applies the same demand system within each sector of the multi-sector environment in Section 3. Section D shows that the gains-from-trade formula is unchanged when the Armington goods are replaced by Krugman varieties with free entry. Section E introduces heterogeneous firms and Pareto productivity and derives the corresponding welfare-relevant elasticity function.

C.1 Environment

Except for the demand system, the environment and notation are the same as in Section 2.¹⁶

Notation. For later use, let $E_n \equiv \sum_{i \in \mathcal{N}} X_{ni} = Y_n$ denote country n 's total expenditure and, for each foreign source $i \neq n$, let $\mu_{ni} \equiv X_{ni}/X_{nF}$ denote its share of country n 's foreign expenditure. Thus, $X_{ni}/E_n = (1 - \lambda_{nn})\mu_{ni}$.

Preferences. We use a nested homothetic CDE-CES expenditure system: a CDE upper nest over home goods and an aggregation of foreign goods, and a constant-elasticity-of-substitution (CES) foreign-source aggregator. Because preferences are homothetic, they can be represented by an implicit unit expenditure function.¹⁷

The unit expenditure function $e_n(\mathbf{p}_n)$ is defined implicitly by

$$\omega_{nn} \left(\frac{e_n}{p_{nn}} \right)^{\theta_0} + \left[\sum_{i \neq n} \omega_{ni} \left(\frac{e_n}{p_{ni}} \right)^{\rho} \right]^{(\theta_0 + \theta_1)/\rho} = 1, \quad (25)$$

where $\theta_0 > 0$, $\theta_0 + \theta_1 > 0$, $\rho > 0$, $\omega_{nn} > 0$, and $\omega_{ni} > 0$. The parameters θ_0 , θ_1 , and ρ shape substitution elasticities, as shown below. The parameters ω_{ni} shape relative expenditure

¹⁶The Armington formulation is not essential; see Sections D and E.

¹⁷We use a CES lower nest for simplicity only. A single-nested CDE model would already be strictly more general than the canonical quantitative trade models.

shares across foreign sources, while ω_{nn} shifts demand for the home good relative to the foreign aggregation. When $\theta_1 = 0$, the elasticity between the home aggregate and the foreign aggregate is constant and equal to θ_0 . When, in addition, $\rho = \theta_0$, equation (25) is equivalent to the standard CES unit expenditure function/price index, where the elasticity of substitution is $1 + \theta_0$.

C.2 Characterization

Define $A_{nn} \equiv \omega_{nn} (e_n/p_{nn})^{\theta_0}$ and $A_{nF} \equiv \left[\sum_{i \neq n} \omega_{ni} (e_n/p_{ni})^\rho \right]^{(\theta_0 + \theta_1)/\rho}$. According to equation (25),

$$A_{nn} + A_{nF} = 1. \quad (26)$$

Let

$$B_n \equiv \theta_0 A_{nn} + (\theta_0 + \theta_1) A_{nF}.$$

Holding preference weights fixed and differentiating equation (26) with respect to prices gives

$$\theta_0 A_{nn} (d \ln e_n - d \ln p_{nn}) + (\theta_0 + \theta_1) A_{nF} \left(d \ln e_n - \sum_{i \neq n} \mu_{ni} d \ln p_{ni} \right) = 0,$$

where

$$\mu_{ni} \equiv \frac{\omega_{ni} p_{ni}^{-\rho}}{\sum_{k \neq n} \omega_{nk} p_{nk}^{-\rho}}.$$

Solving the previous expression for $d \ln e_n$ and combining with Shephard's lemma yields the domestic expenditure share

$$\lambda_{nn} = \frac{X_{nn}}{E_n} = \frac{\partial \ln e_n}{\partial \ln p_{nn}} = \frac{\theta_0 A_{nn}}{B_n}, \quad (27)$$

and, for each foreign source $i \neq n$,

$$\lambda_{ni} = \frac{X_{ni}}{E_n} = \frac{\partial \ln e_n}{\partial \ln p_{ni}} = \frac{(\theta_0 + \theta_1) A_{nF}}{B_n} \mu_{ni}. \quad (28)$$

The restricted CDE system in equation (25) can be written as a CDE system across two aggregates,

$$\omega_{nn} \left(\frac{e_n}{p_{nn}} \right)^{\theta_0} + \left(\frac{e_n}{P_{nF}} \right)^{\theta_0 + \theta_1} = 1, \quad (29)$$

given the following definition

$$P_{nF} \equiv \left(\sum_{i \neq n} \omega_{ni} p_{ni}^{-\rho} \right)^{-1/\rho}.$$

The restriction in equation (25) is that the many foreign sources enter the foreign aggregate with the common exponent ρ . At the upper level, however, equation (29) is an unrestricted two-good CDE system, up to the normalization of the foreign-aggregator's weight.

The upper-level exponents are θ_0 and $\theta_0 + \theta_1$, while ρ governs substitution across foreign sources inside the foreign aggregate. The share of foreign expenditure on source country i is

$$\mu_{ni} \equiv \frac{X_{ni}}{X_{nF}} = \frac{\omega_{ni} p_{ni}^{-\rho}}{\sum_{k \neq n} \omega_{nk} p_{nk}^{-\rho}}.$$

Therefore, for any two foreign source countries $i, j \neq n$,

$$\frac{X_{ni}}{X_{nj}} = \frac{\omega_{ni}}{\omega_{nj}} \left(\frac{p_{ni}}{p_{nj}} \right)^{-\rho}.$$

Relative expenditure across foreign source countries is CES, and the across-foreign-source trade elasticity is ρ .

According to equations (27) and (28),

$$\frac{X_{nn}}{X_{nF}} = \frac{\lambda_{nn}}{1 - \lambda_{nn}} = \frac{\theta_0 A_{nn}}{(\theta_0 + \theta_1) A_{nF}}.$$

Log differentiating the previous expression holding all ω terms fixed,

$$d \ln \frac{X_{nn}}{X_{nF}} = d \ln A_{nn} - d \ln A_{nF},$$

where, again holding all ω terms fixed, we have

$$d \ln A_{nn} = \theta_0 (d \ln e_n - d \ln p_{nn}),$$

and

$$d \ln A_{nF} = (\theta_0 + \theta_1) (d \ln e_n - d \ln P_{nF}).$$

Using Shephard's lemma, we have

$$d \ln e_n = \lambda_{nn} d \ln p_{nn} + (1 - \lambda_{nn}) d \ln P_{nF}.$$

The previous expressions therefore yield

$$d \ln \frac{X_{nn}}{X_{nF}} = (\theta_0 + \theta_1 \lambda_{nn}) d \ln \left(\frac{P_{nF}}{p_{nn}} \right).$$

The previous expression implies

$$\varepsilon_n(\lambda_{nn}) \equiv \frac{\partial \ln(X_{nn}/X_{nF})}{\partial \ln(P_{nF}/p_{nn})} = \theta_0 + \theta_1 \lambda_{nn}.$$

Discussion. The three parameters in the unit expenditure function in equation (25) have distinct roles. First, the across-foreign-source trade elasticity is ρ . Second, the value of the home-foreign trade elasticity at $\lambda_{nn} = 0$ is θ_0 . Third, the slope of the home-foreign trade elasticity in the domestic expenditure share λ_{nn} is θ_1 . Thus the model allows the across-foreign elasticity, the slope of the home-foreign elasticity, and the gap between the two elasticities at a given domestic expenditure share to vary separately.

Gains from Trade. Substituting the linear elasticity function into Proposition 1 gives the following closed form for two equilibria, indexed by 0 and 1, across which country n 's productivity, labor, and preference weights are fixed:

$$\ln \frac{W_n^0}{W_n^1} = \frac{1}{\theta_0} \ln \left[\frac{\lambda_{nn}^1 (\theta_0 + \theta_1 \lambda_{nn}^0)}{\lambda_{nn}^0 (\theta_0 + \theta_1 \lambda_{nn}^1)} \right].$$

If equilibrium 0 is the trade equilibrium, indicated with T , and equilibrium 1 is autarky, then $\lambda_{nn}^1 = 1$ and gains from trade are

$$\ln \frac{W_n^T}{W_n^A} = \frac{1}{\theta_0} \ln \left[\frac{\theta_0 + \theta_1 \lambda_{nn}^T}{(\theta_0 + \theta_1) \lambda_{nn}^T} \right]. \quad (30)$$

As $\theta_1 \rightarrow 0$, equation (30) converges to the CES/ACR formula (Arkolakis et al., 2012)

$$\lim_{\theta_1 \rightarrow 0} \ln \frac{W_n^T}{W_n^A} = \frac{-1}{\theta_0} \ln (\lambda_{nn}^T). \quad (31)$$

The elasticity θ_0 in equation (31) is the elasticity between home expenditure and aggregate foreign expenditure. The across-foreign elasticity ρ does not enter the gains formula. When $\theta_1 = 0$ and $\rho = \theta_0$, the home-foreign elasticity equals the elasticity across foreign sources and using ρ for welfare is therefore correct.

C.3 The Multi-Sector Version

The environment and notation are those of Section 3.1. We impose Cobb-Douglas aggregation across sectors,

$$\mathcal{E}_n((e_{ns})_{s \in \mathcal{S}}) = \prod_{s \in \mathcal{S}} \left(\frac{e_{ns}}{\alpha_{ns}} \right)^{\alpha_{ns}},$$

where $\alpha_{ns} > 0$ and $\sum_{s \in \mathcal{S}} \alpha_{ns} = 1$, and use the CDE-CES expenditure system from Section C within each sector:

$$\omega_{nns} \left(\frac{e_{ns}}{p_{nns}} \right)^{\theta_{s0}} + \left[\sum_{i \neq n} \omega_{nis} \left(\frac{e_{ns}}{p_{nis}} \right)^{\rho_{ns}} \right]^{(\theta_{s0} + \theta_{g(s)1})/\rho_{ns}} = 1,$$

where $\theta_{s0} > 0$, $\theta_{s0} + \theta_{g(s)1} > 0$, $\rho_{ns} > 0$, and all preference weights are positive. As in the empirical specification, $g(s)$ denotes the sectoral aggregate containing sector s .

The derivations above apply separately within each sector. In particular, relative expenditure across foreign sources is CES:

$$\frac{X_{nis}}{X_{njs}} = \frac{\omega_{nis}}{\omega_{njs}} \left(\frac{p_{nis}}{p_{njs}} \right)^{-\rho_{ns}} \quad \text{for all } i, j \neq n,$$

while the sectoral home-foreign elasticity is

$$\varepsilon_{ns}(\lambda_{nns}) = \theta_{s0} + \theta_{g(s)1} \lambda_{nns}.$$

Thus ρ_{ns} is the elasticity across foreign sources, θ_{s0} is the intercept of the home-foreign elasticity function, and $\theta_{g(s)1}$ is its slope in the domestic expenditure share. Under Cobb-Douglas aggregation, substituting this linear elasticity function into equation (12) gives

$$\ln \frac{W_n^T}{W_n^A} = \sum_{s \in S} \frac{\alpha_{ns}}{\theta_{s0}} \ln \left[\frac{\theta_{s0} + \theta_{g(s)1} \lambda_{nns}^T}{(\theta_{s0} + \theta_{g(s)1}) \lambda_{nns}^T} \right].$$

D A Krugman Gravity Model

We now replace the Armington good produced by each country with an endogenous mass of differentiated firm varieties, as in [Krugman \(1980\)](#). We retain the CDE upper tier between the domestic and foreign aggregates and CES demand with exponent ρ across firm varieties. We show that the resulting gains-from-trade formula is exactly the same as in the Armington model.

Origin i has a mass M_i of symmetric firms. Each firm has marginal cost $c_i = w_i/a_i$, and entry requires f_i^e units of country i 's factor. Delivering a variety from country i to country n requires the iceberg trade cost $\mathcal{D}_{ni}$, with $\mathcal{D}_{nn} = 1$. Because the elasticity of substitution across firm varieties is $1 + \rho$, the consumer price of each variety is

$$q_{ni} = \frac{1 + \rho}{\rho} c_i \mathcal{D}_{ni}.$$

Let $\zeta_{ni} > 0$ denote a fixed destination-origin demand shifter. The price index for the composite of country i 's varieties sold in country n is

$$P_{ni} = [M_i \zeta_{ni}^\rho q_{ni}^{-\rho}]^{-1/\rho}.$$

The domestic and foreign price indexes are therefore

$$P_{nn} = [M_n \zeta_{nn}^\rho q_{nn}^{-\rho}]^{-1/\rho}$$

and

$$P_{nF} = \left[\sum_{i \neq n} M_i \zeta_{ni}^\rho q_{ni}^{-\rho} \right]^{-1/\rho}.$$

Country n 's unit expenditure function is defined by the same CDE upper tier as in equation (29):

$$\omega_{nn} \left(\frac{e_n}{P_{nn}} \right)^{\theta_0} + \left(\frac{e_n}{P_{nF}} \right)^{\theta_0 + \theta_1} = 1.$$

This upper tier implies that the elasticity between domestic and foreign expenditure varies with the domestic expenditure share. However, within the CES variety aggregate, each atomistic firm faces the constant demand elasticity $1 + \rho$ and therefore charges the same constant markup $(1 + \rho)/\rho$ as in the standard CES Krugman model.

A constant markup, the free-entry condition, and factor-market clearing together imply that the mass of firms is independent of trade costs and expenditure patterns, as in the

standard (CES) Krugman model. Let $\bar{R}_i$ denote the total revenue of a representative country- i firm across all destinations. Operating profits are $\bar{R}_i/(1 + \rho)$, so free entry implies

$$\frac{\bar{R}_i}{1 + \rho} = w_i f_i^e.$$

The variable factor bill is the fraction $\rho/(1 + \rho)$ of revenue and therefore equals $\rho w_i f_i^e$. Including the entry cost, each firm uses $(1 + \rho)f_i^e$ units of the factor. Factor-market clearing then gives

$$M_i = \frac{L_i}{(1 + \rho)f_i^e}. \quad (32)$$

Thus M_i depends on the factor endowment and entry cost but not on trade costs or expenditure patterns.

For country n , the domestic price index can be written as

$$P_{nn} = \frac{1 + \rho}{\rho} \frac{w_n}{a_n} \frac{1}{\zeta_{nn} M_n^{1/\rho}}.$$

Since $Y_n = w_n L_n$,

$$\frac{Y_n}{P_{nn}} = \frac{\rho}{1 + \rho} a_n L_n \zeta_{nn} M_n^{1/\rho}.$$

The previous expression implies that Y_n/P_{nn} is unchanged across equilibria for fixed values of technology, a_n , factor endowments, L_n , demand shifters, ζ_{nn} , and entry costs. The same cancellation between nominal income and the domestic price is used to derive the Armington welfare formula.

Let λ_{nn} now denote the share of country n 's expenditure allocated to the domestic firm composite. The derivation in Section C.2 applies directly to the CDE upper tier:

$$d \ln \left(\frac{e_n}{P_{nn}} \right) = \frac{d \ln \lambda_{nn}}{\theta_0 + \theta_1 \lambda_{nn}}.$$

More generally, before imposing equation (32), the welfare comparison between equilibria 0 and 1 would be

$$\ln \frac{W_n^0}{W_n^1} = \int_{\lambda_{nn}^0}^{\lambda_{nn}^1} \frac{d \ln \lambda}{\theta_0 + \theta_1 \lambda} + \frac{1}{\rho} \ln \frac{M_n^0}{M_n^1}.$$

Equation (32) makes the second term zero. Therefore, if equilibrium 0 is the trade equilibrium and equilibrium 1 is autarky,

$$\ln \frac{W_n^T}{W_n^A} = \int_{\lambda_{nn}^T}^1 \frac{d \ln \lambda}{\theta_0 + \theta_1 \lambda} = \frac{1}{\theta_0} \ln \left[\frac{\theta_0 + \theta_1 \lambda_{nn}^T}{(\theta_0 + \theta_1) \lambda_{nn}^T} \right].$$

This is exactly the Armington gains-from-trade formula in equation (30). When $\theta_1 = 0$, it reduces to the canonical Krugman result in Arkolakis et al. (2012).

E A Melitz-Pareto Gravity Model

We next replace the symmetric firms in Section D with heterogeneous firms and endogenous selection into each destination, as in Melitz (2003), imposing Pareto distributed productivities. We retain the CDE upper tier between the domestic and foreign aggregates and CES demand with exponent ρ across firm varieties. This extension nests the constant-elasticity welfare result in Feenstra et al. (2018) and shows how that result changes when the home-foreign elasticity varies with the domestic expenditure share.

Each origin has a fixed mass of potential firms, normalized to one. A firm in country i draws productivity φ from a Pareto distribution with $\Pr(\Phi \geq \varphi) = \varphi^{-\kappa}$ for $\varphi \geq 1$, where $\kappa > \rho$. Serving country n requires f_{ni} units of country i 's factor, in addition to the iceberg trade cost $\mathcal{D}_{ni}$. Let φ_{ni}^* denote the productivity cutoff for serving country n and let $N_{ni} = (\varphi_{ni}^*)^{-\kappa}$ denote the mass of active country- i firms in that market. We assume selection is active throughout the path to autarky. Specifically, we assume that every productivity cutoff for market entry remains strictly above the Pareto lower support, $\varphi_{ni}^* > 1$. Since the elasticity of substitution across firm varieties is $1 + \rho$, firms charge the constant markup $(1 + \rho)/\rho$. Integrating over the active firms gives the price index

$$P_{ni} = A_{ni} c_i \mathcal{D}_{ni} N_{ni}^{1/\kappa - 1/\rho},$$

where $A_{ni} > 0$ collects the markup, Pareto, and fixed demand-shifter terms. The zero-profit condition for the cutoff firm and the Pareto distribution also imply

$$X_{ni} = \frac{(1 + \rho)\kappa}{\kappa - \rho} w_i f_{ni} N_{ni}.$$

Thus, changes in the mass of firms serving a market can be recovered from changes in expenditure and the origin wage.

Country n 's unit expenditure function is defined by the same CDE upper tier as in equation (29), replacing the Armington prices with the domestic and foreign firm-variety price indexes. Let

$$\varepsilon_n^{HF}(\lambda) \equiv \theta_0 + \theta_1 \lambda$$

denote the home-foreign expenditure elasticity generated by this upper tier. Consumers own the domestic firms, so Y_n includes factor income and firm profits. In the one-sector model, combining factor-market clearing and the balanced-budget condition yields

$$Y_n = \frac{1 + \rho}{1 + \rho - \rho/\kappa} w_n L_n.$$

The coefficient multiplying factor income is constant across equilibria. Hence, $d \ln Y_n = d \ln w_n$. Since $X_{nn} = \lambda_{nn} Y_n$, the preceding expenditure equation implies

$$d \ln N_{nn} = d \ln X_{nn} - d \ln w_n = d \ln \lambda_{nn}.$$

The domestic firm-variety price index therefore satisfies

$$d \ln P_{nn} = d \ln w_n + \left(\frac{1}{\kappa} - \frac{1}{\rho} \right) d \ln \lambda_{nn},$$

and thus

$$d \ln \left(\frac{Y_n}{P_{nn}} \right) = \left(\frac{1}{\rho} - \frac{1}{\kappa} \right) d \ln \lambda_{nn}.$$

The CDE upper tier implies

$$d \ln \left(\frac{e_n}{P_{nn}} \right) = \frac{d \ln \lambda_{nn}}{\varepsilon_n^{HF}(\lambda_{nn})}.$$

Using $W_n = Y_n/e_n$ and combining the previous two expressions gives

$$d \ln W_n = - \left[\frac{1}{\varepsilon_n^{HF}(\lambda_{nn})} + \frac{1}{\kappa} - \frac{1}{\rho} \right] d \ln \lambda_{nn}.$$

It is therefore useful to define the welfare-relevant Melitz-Pareto elasticity function by

$$\frac{1}{\varepsilon_n^{MP}(\lambda)} \equiv \frac{1}{\theta_0 + \theta_1 \lambda} + \frac{1}{\kappa} - \frac{1}{\rho}. \quad (33)$$

Provided this expression is positive and the interior-selection restriction above holds along the path to autarky, gains from trade are

$$\begin{aligned} \ln \frac{W_n^T}{W_n^A} &= \int_{\lambda_{nn}^T}^1 \frac{d \ln \lambda}{\varepsilon_n^{MP}(\lambda)} \\ &= \frac{1}{\theta_0} \ln \left[\frac{\theta_0 + \theta_1 \lambda_{nn}^T}{(\theta_0 + \theta_1) \lambda_{nn}^T} \right] + \left(\frac{1}{\kappa} - \frac{1}{\rho} \right) \ln \left(\frac{1}{\lambda_{nn}^T} \right). \end{aligned}$$

The first term is the Armington and Krugman expression in equation (30); the second term captures firm selection.

To relate equation (33) to Feenstra et al. (2018), set $\theta_1 = 0$, let $\omega = 1 + \theta_0$ denote their macro elasticity of substitution, and let $\sigma = 1 + \rho$ denote their micro elasticity of substitution. The welfare-relevant elasticity is then constant and can be written

$$\varepsilon_n^{MP} = \frac{\kappa(\omega - 1)}{(\sigma - \omega)(\kappa/(\sigma - 1) - 1) + \sigma - 1},$$

which is their trade-elasticity formula. When $\sigma = \omega$, or equivalently $\rho = \theta_0$, this elasticity equals the Pareto shape parameter κ , as in the standard Melitz-Pareto model. Equation (33) therefore replaces their constant macro elasticity with the linear home-foreign elasticity function while retaining their firm-selection mechanism.

Quantitative Appendix

F Intermediate-Input Multiplier

This section derives equation (24). Let R_n denote the increase in country n 's aggregate unit expenditure moving from trade to autarky while holding all domestic sectoral unit costs

fixed. Under the nested CES system in Section 5.2,

$$\ln R_n = \frac{1}{1 - \sigma_U} \ln \left[\alpha_{nM} \hat{E}_{nM}^{1-\sigma_U} + \alpha_{nG} \hat{E}_{nG}^{1-\sigma_U} + \alpha_{n0} \right].$$

Let $\hat{P}_n \equiv P_n^A / P_n^T$, $\hat{w}_n \equiv w_n^A / w_n^T$, and $\hat{c}_n$ denote the common proportional change in domestic sectoral unit costs. Homogeneity of the expenditure system implies

$$\hat{P}_n = \hat{c}_n R_n.$$

Because every sector has the same value-added share and uses the same intermediate composite, its unit-cost change is

$$\hat{c}_n = \hat{w}_n^{\beta_n} \hat{P}_n^{1-\beta_n}.$$

Combining these equations gives

$$\left(\frac{\hat{P}_n}{\hat{w}_n} \right)^{\beta_n} = R_n.$$

Fixed factor endowments imply that nominal income changes in proportion to the wage. Hence,

$$\frac{W_n^T}{W_n^A} = \frac{\hat{P}_n}{\hat{w}_n} = R_n^{1/\beta_n},$$

which yields equation (24). When $\beta_n = 1$, the formula reduces to the no-intermediate-input expression in equation (11).

Sensitivity to the Positivity Bound. Table 19 repeats the quantification after lowering the bound on each elasticity function from 0.1 to 0.05 and re-estimating finite-change Specification (3). Median gains change little, while mean gains rise from 32.00 to 40.55 log points in WIOD and from 26.90 to 33.44 in OECD. Holding each country-sector elasticity fixed at its initial value continues to produce larger mean gains under both bounds.

Table 19: Gains from Trade Sensitivity to the Elasticity Lower Bound

Data	Lower bound	Variable elasticities		Fixed initial elasticities	
		Mean	Median	Mean	Median
WIOD	0.10	32.00	19.37	54.42	24.43
WIOD	0.05	40.55	19.54	94.56	24.60
OECD	0.10	26.90	19.59	50.46	21.21
OECD	0.05	33.44	19.98	84.31	22.02

Notes: This table repeats the gains-from-trade calculations using a lower bound of 0.05 on each elasticity function, compared with 0.1 in the baseline. We re-estimate finite-change Specification (3) under each bound. Both calculations use $\sigma_L = 2$, $\sigma_U = 0.6$, and the same 2007 expenditure shares and value-added shares. Fixed initial elasticities hold each country-sector elasticity at its fitted 2007 value under the corresponding bound. Gains are in log points. The WIOD and OECD samples contain 40 and 80 countries, respectively. We report point estimates only.

G OECD ICIO Quantification

This appendix repeats the WIOD quantification but now using OECD ICIO. We use the 2007 trade equilibrium, 19 manufacturing sectors and seven non-manufacturing goods sectors, and treat the remaining non-goods sectors as a single non-traded composite. We construct the home-foreign elasticity functions from the constrained finite-change estimates in Table 14. We calculate each country's value-added share from OECD ICIO value added and gross output; see Section 5.1. As in the baseline exercise, the subsequent results use Specification (3), $\sigma_L = 2$, and $\sigma_U = 0.6$ unless otherwise stated.

Table 20: OECD ICIO Gains from Trade Across Elasticity-function Specifications

	p10	p25	Median	Mean	p75	p90
(1)	8.19	10.73	16.72	20.69	25.77	37.80
(2)	14.62	17.98	25.12	33.31	35.22	54.95
(3)	10.96	13.50	19.59	26.90	27.48	44.00

Notes: Entries report $100 \ln(W_n^T/W_n^A)$ calculated using equation (24), where β_n is 2007 economy-wide value added divided by gross output. The percentiles and mean are calculated across the 80 countries in the OECD ICIO quantification sample. All three specifications use the finite-change, positive-floor spline estimates in Table 14 and set $\sigma_L = 2$ and $\sigma_U = 0.6$.

Table 20 reports the distribution across the 80 OECD ICIO countries. Under Specification (3), the median and mean gains from trade are 19.59 and 26.90 log points, compared with 19.37 and 32.00 log points in WIOD. The mean ranges from 20.69 to 33.31 log points across Specifications (1)-(3).

The median gains are also similar on a common country sample. Among the 40 countries common to the two quantifications, the median gain from trade is 20.17 log points using OECD ICIO and 19.37 log points using WIOD. This similarity holds despite differences in trade exposure and estimated elasticity functions. Among these countries, the median share of total expenditure devoted to goods imports is 12.3 percent in OECD ICIO and 15.7 percent in WIOD, while the median share devoted to goods-producing sectors is 34.1 and 35.5 percent. Under Specification (3), the average fitted elasticities at the sector-specific median domestic expenditure shares among these countries are 3.92 and 3.92 in OECD ICIO for consumer and producer manufacturing, compared with 3.44 and 8.28 in WIOD; the corresponding non-manufacturing-goods values are 8.37 and 1.05. The median value-added share among these countries is 0.47 in both data sets. Thus, the two data sets yield similar median gains from trade despite differences in measured trade exposure and estimated home-foreign elasticity functions.

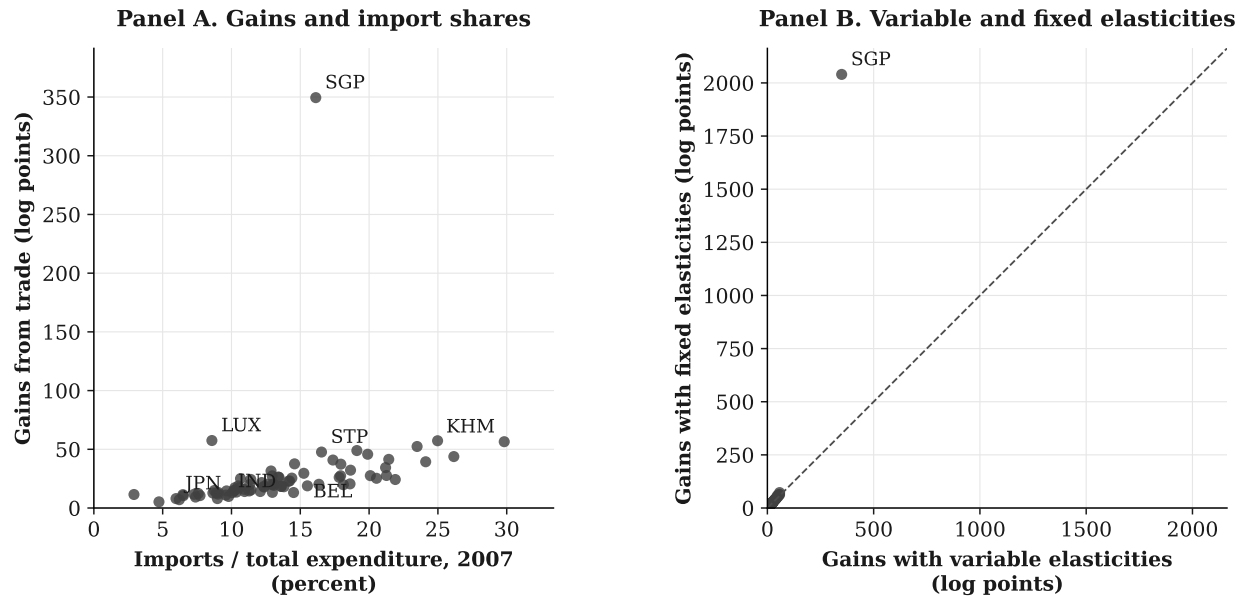


Figure 5: OECD ICIO Gains from Trade, Import Exposure, and Variable Elasticities

Notes: Each observation is one of the 80 countries in the OECD ICIO quantification sample. Both panels use estimates from Specification (3) in Table 14, with $\sigma_L = 2$, $\sigma_U = 0.6$, and the country-specific intermediate-input multiplier $1/\beta_n$. Gains from trade are measured as $100 \ln(W_n^T/W_n^A)$. Panel A plots gains against 2007 expenditure on imports in the 26 goods-producing sectors divided by total expenditure, including the non-traded composite. Panel B plots gains holding each country-sector elasticity fixed at its fitted 2007 value against gains under the variable-elasticity functions. The dashed line is a 45-degree line. Selected countries are labeled.

Figure 5 repeats the main figure. Holding each country-sector elasticity fixed at its 2007 value raises the median gain from 19.59 to 21.21 log points and raises gains for every country.

Singapore is an extreme outlier in both Panel A and B. Its gains from trade are 349.48 log points under the variable-elasticity specification and 2,040.04 log points when elasticities are held fixed at their initial values. These large gains primarily reflect the non-manufacturing composite: several sectors have zero domestic expenditure, while agriculture and fishing have low initial fitted home-foreign elasticities. Holding these elasticities fixed makes substitution toward domestic goods along the path to autarky particularly costly. Singapore's low value-added share further amplifies the gains through intermediate inputs. Excluding Singapore, the mean gains under the variable-elasticity specification fall from 26.90 to 22.81 log points; of course, median gains are less sensitive.

Table 21 varies the cross-sector elasticities. Lowering σ_L to one raises the median to 55.21 log points, while raising it to three lowers the median to 15.14 log points. As in WIOD, zero domestic expenditure in some sectors implies infinite gains for some countries when $\sigma_L = 1$. Setting $\sigma_U = 1$ lowers the median to 18.59 log points.

Table 21: OECD ICIO Gains from Trade Sensitivity to Cross-Sector Substitution

σ_L	σ_U	p10	p25	Median	Mean	p75	p90
2	0.6	10.96	13.50	19.59	26.90	27.48	44.00
1	0.6	17.90	28.27	55.21	∞	∞	∞
3	0.6	8.23	10.74	15.14	19.77	21.63	33.51
2	1	10.33	13.17	18.59	22.23	26.53	40.69

Notes: Entries report $100 \ln(W_n^T/W_n^A)$ calculated using equation (24), where β_n is 2007 economy-wide value added divided by gross output. The percentiles and mean are calculated across the 80 countries in the OECD ICIO quantification sample. Specification (3) in Table 14 is used throughout. The baseline sets $\sigma_L = 2$ and $\sigma_U = 0.6$; each subsequent row changes the elasticities shown.

H Details of Monte Carlo Exercises

This appendix uses model-generated data to ask whether the estimator generates positive average slopes when the true functions are constant and whether it recovers positive slopes when the true functions are linear. We use the quantitative multi-sector Armington model developed in Appendix C. The calibrations share the same initial equilibrium and tariff shocks. Within each calibration, the measured-price-control intercept varies by sector, while the additive-control intercept varies by group. The estimates reported here use the direct-price specification in Section 4.

We simulate an economy with 40 countries and 16 goods sectors, matching the dimensions of the WIOD baseline. The 16 sectors consist of 14 manufacturing sectors and two non-manufacturing sectors. As in WIOD, manufacturing is divided evenly into seven consumer and seven producer sectors. Preferences are Cobb-Douglas across sectors. Within each sector, demand follows the CDE-CES system

$$\omega_{nns} \left(\frac{e_{ns}}{p_{nns}} \right)^{\theta_{s0}} + \left[\sum_{i \neq n} \omega_{nis} \left(\frac{e_{ns}}{p_{nis}} \right)^{\rho_s} \right]^{(\theta_{s0} + \theta_{s1})/\rho_s} = 1.$$

As shown in Appendix C, the elasticity across foreign sources is ρ_s , while the home-foreign elasticity is $\varepsilon_s(\lambda) = \theta_{s0} + \theta_{s1}\lambda$. Thus, the model permits these two elasticities to differ and generates the linear special case nested in the estimated spline.

We hold labor endowments, Cobb-Douglas sectoral expenditure shares, iceberg trade costs, initial tariffs, and across-foreign-source elasticities fixed throughout the exercise. These objects are drawn once. Log labor endowments are independent normal draws with mean zero and standard deviation 0.40. We first draw common sectoral expenditure weights from a Dirichlet distribution with all 16 parameters equal to three. Conditional on these weights, each country's expenditure shares are an independent Dirichlet draw with parameters equal to 80 times the common weights plus 0.5 in each sector.

We draw each country's location independently and uniformly on the unit square and denote the distance between countries n and i by D_{ni} . Foreign log iceberg costs equal $1.35 + 1.20D_{ni} + \eta_{nis}$, where η_{nis} is an independent normal draw with mean zero and standard deviation 0.10. Domestic iceberg costs equal one. Initial log gross tariffs are independent uniform draws between zero and $\log(1.20)$ for foreign goods; domestic tariffs are zero. In

the initial CES economy, the logs of the origin-sector preference weights are independent normal draws with mean zero and standard deviation 0.45. These weights are common across destinations. We draw $\log \tilde{\rho}_s \sim \mathcal{N}(\log 2, 0.35^2)$ independently across sectors and set $\rho_s = \min\{4.5, \max\{0.9, \tilde{\rho}_s\}\}$. We impose $\theta_{s0} = \rho_s$ throughout.

We consider four calibrations. Calibration 1 sets $\theta_{s1} = 0$ in every sector, yielding CES demand across all origins. Calibration 2 sets $\theta_{s1} = 4$ in every sector. Calibration 3 sets $\theta_{s1} = 3$ in every manufacturing sector and $\theta_{s1} = 6$ in both non-manufacturing sectors. Calibration 4 sets $\theta_{s1} = 3$ in every consumer-manufacturing sector, $\theta_{s1} = 5$ in every producer-manufacturing sector, and $\theta_{s1} = 7$ in both non-manufacturing sectors. Thus, the pooling imposed by Specification (2) matches Calibration 3, while the pooling imposed by Specification (3) matches Calibration 4.

We first solve Calibration 1 for its initial equilibrium wages and bilateral expenditure shares. Initial CES expenditure shares are proportional to the origin-sector preference weight times the delivered price raised to $-\rho_s$, where the delivered price is the origin's wage times the iceberg cost and gross tariff. We treat tariffs as trade-cost wedges and do not model tariff revenue. Expenditure equals labor income, and wages equate each country's labor income to its sales. We normalize the geometric mean wage to one. We then choose the preference weights in Calibrations 2–4 so that, at the same wages and tariffs, they reproduce the initial bilateral shares, normalizing each initial country-sector unit expenditure index to one. The four calibrations therefore have the same initial equilibrium and domestic-share distribution. They differ in the functions governing their responses to tariff shocks.

We generate 200 tariff-shock matrices. In each matrix, every foreign destination-origin-sector cell is selected independently with probability 0.45. In each selected cell, the log gross tariff is reduced by a fraction equal to 0.25 times an independent Beta(2, 2) draw; the other tariffs are unchanged. We apply the same matrices to all four calibrations and solve for the new equilibrium wages and expenditure shares, giving 800 equilibrium changes. We use NumPy's PCG64 random-number generator, with seed 20260930 for the initial economy and 71011 for the tariff shocks. The reported distributions therefore condition on one initial economy and vary only the tariff shocks.

For each change, we construct the dependent variable, initial domestic shares, tariff shifter, and measured-price control. Unit-cost changes equal the supplying country's wage change in this model. We observe those changes in the simulation and use them to construct Q_{ns}^P , with initial foreign shares and the domestic-price subtraction. The randomized tariff shifter instruments itself. We estimate the unrestricted three-knot Specifications (1)–(3), which pool all goods, separate manufacturing from non-manufacturing, or additionally separate consumer from producer manufacturing. Elasticity and measured-price-control intercepts remain sector-specific, while the additive-control intercept varies by group.

The tables report the mean and the 5th and 95th percentiles across the 200 draws of the derivative evaluated at each sector's median initial share and then averaged across sectors. When slopes differ across sectors within a sectoral group, we report the average value. The reported percentiles summarize how the estimates vary across simulated tariff changes.

Calibration 1: Constant Home-Foreign Elasticities. Table 22 reports the estimates when all true slopes are zero. The mean estimated all-goods slope at sector medians is 0.002,

and the group estimates are also close to zero. Thus, the estimator does not mechanically produce positive average slopes in this constant-elasticity environment.

Table 22: Monte Carlo Estimates with Constant Elasticities

Aggregate	True mean	Est. mean	5th–95th pct.	Share > 0
All goods	0.000	0.002	[-0.004, 0.008]	67.5%
Manufacturing	0.000	0.001	[-0.004, 0.008]	60.5%
Consumer manuf.	0.000	0.003	[-0.006, 0.012]	66.0%
Producer manuf.	0.000	-0.001	[-0.012, 0.011]	45.5%
Non-manuf. goods	0.000	0.000	[-0.028, 0.031]	49.0%

Notes: Each row reports results from 200 simulations using the unrestricted three-knot spline. In each simulation, we evaluate the estimated slope at each sector’s median initial domestic expenditure share and average equally across sectors in the row. “Est. mean” averages these estimates across the 200 simulations. “True mean” is the average of the true slopes across sectors in the row. The reported percentiles summarize how the estimates vary across simulations. “Share > 0” is the fraction of simulations with a positive average estimated slope. The model has 40 countries and 16 goods sectors: seven consumer-manufacturing, seven producer-manufacturing, and two non-manufacturing sectors. We use the same tariff shocks across calibrations in each simulation.

Calibration 2: A Common Positive Slope. Table 23 sets the true slope to four in every sector. The all-goods mean estimate at sector medians is 4.03. The manufacturing, consumer-manufacturing, producer-manufacturing, and non-manufacturing estimates also closely recover the common true slope.

Table 23: Monte Carlo Estimates with a Common True Slope of Four

Aggregate	True mean	Est. mean	5th–95th pct.	Share > 0
All goods	4.000	4.027	[4.015, 4.039]	100.0%
Manufacturing	4.000	4.024	[4.011, 4.038]	100.0%
Consumer manuf.	4.000	4.017	[4.002, 4.036]	100.0%
Producer manuf.	4.000	4.032	[4.008, 4.056]	100.0%
Non-manuf. goods	4.000	4.050	[3.989, 4.119]	100.0%

Notes: This table replicates Table 22, but the true slope is four in every sector.

Calibration 3: Manufacturing and Non-Manufacturing Slopes. Table 24 reports estimates when the true slope is three in manufacturing and six in non-manufacturing. The mean estimates for these groups are 3.02 and 6.07. The all-goods specification pools sectors with different slopes and yields a mean estimate of 3.23.

Table 24: Monte Carlo Estimates with Two Group-Specific Slopes

Aggregate	True mean	Est. mean	5th–95th pct.	Share > 0
All goods	3.375	3.230	[3.158, 3.317]	100.0%
Manufacturing	3.000	3.021	[3.011, 3.033]	100.0%
Consumer manuf.	3.000	3.017	[3.004, 3.033]	100.0%
Producer manuf.	3.000	3.026	[3.006, 3.047]	100.0%
Non-manuf. goods	6.000	6.070	[5.994, 6.157]	100.0%

Notes: This table replicates Table 22, but the true slopes are three in manufacturing and six in non-manufacturing.

Calibration 4: Consumer-Manufacturing, Producer-Manufacturing, and Non-Manufacturing Slopes. Table 25 reports estimates when the true slopes are three in consumer manufacturing, five in producer manufacturing, and seven in non-manufacturing. The corresponding mean estimates are 3.02, 5.04, and 7.08.

Table 25: Monte Carlo Estimates with Three Group-Specific Slopes

Aggregate	True mean	Est. mean	5th–95th pct.	Share > 0
All goods	4.375	4.140	[3.984, 4.307]	100.0%
Manufacturing	4.000	3.916	[3.769, 4.081]	100.0%
Consumer manuf.	3.000	3.017	[3.004, 3.033]	100.0%
Producer manuf.	5.000	5.035	[5.008, 5.062]	100.0%
Non-manuf. goods	7.000	7.081	[6.993, 7.178]	100.0%

Notes: This table replicates Table 22, but the true slopes are three in consumer manufacturing, five in producer manufacturing, and seven in non-manufacturing.